

February 23, 2026

Assistant Secretary for Technology Policy
The Office of the National Coordinator for Health Information Technology
U.S. Department of Health and Human Services
Mary E. Switzer Building
7033A, 330 C Street SW, Washington, DC 20201

Re: RFI: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

To Whom It May Concern:

The Health AI Partnership (HAIP) appreciates the opportunity to respond to this RFI. HAIP is a national multi-stakeholder community of practice comprising 35+ healthcare delivery organizations (HDOs) and ecosystem partners, launched in April 2022 as an initiative of the Duke Institute for Health Innovation (DIHI) and Duke University School of Medicine. Our mission is to empower healthcare professionals to use AI effectively, safely, and ethically through community-informed, up-to-date best practices.

HAIP occupies a distinctive position in the healthcare AI landscape. We are a working collaborative that develops freely available governance tools, tests them across diverse healthcare settings, and generates peer-reviewed evidence on what it takes to adopt AI responsibly. Our members span academic medical centers, integrated health systems, and community hospitals, alongside ecosystem partners including the ONC/ASTP, DLA Piper, and Roche. What distinguishes HAIP is its combination of AI implementation support for both high- and low-resource environments, best-practice development shaped by a multidisciplinary stakeholder team, amplification of HDO voices from safety-net organizations to academic medical centers, and policy advocacy that raises awareness for local capacity building.

This response draws on HAIP's published research and implementation experience. About 50 external partners with experience implementing AI in healthcare delivery, industry, legal, and academic settings contributed input that shaped the final response. We welcome the opportunity to discuss these recommendations further and to support the Department's efforts to accelerate responsible AI adoption across the healthcare system.

Sincerely,



Suresh Balu, MS, MBA
Co-Founder, Health AI Partnership
Executive Director,
Duke Institute for Health Innovation



Alifia Hasan, BPharm, MBA
Manager, Health AI Partnership
Innovation Portfolio Manager,
Duke Institute for Health Innovation

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Question 1: Barriers to Innovation

Summary

- **Workforce and capacity:** AI expertise is concentrated in large academic medical centers. Community hospitals, rural facilities, and safety-net providers lack the staff and processes to evaluate, implement, or monitor AI tools, leaving most of the healthcare market unable to participate in AI innovation. Public-private technical assistance programs, modeled on ONC's Regional Extension Centers, could close this gap without expanding government.
- **Data infrastructure:** Fragmented health data and proprietary EHR platforms prevent AI tools validated at one institution from working at another. Stronger interoperability standards and cost-effective pathways for shared computing infrastructure would unlock AI deployment for smaller providers currently priced out of the market.
- **Payment misalignment:** Fee-for-service payment cannot capture AI-driven improvements in outcomes and efficiency, and no reimbursement mechanism covers the cost of adopting AI responsibly, creating a race to the bottom where organizations that skip rigorous evaluation deploy faster and cheaper. Value-based care models offer a more natural alignment but remain far from universal.
- **Regulatory fragmentation:** Overlapping guidance from FDA, ONC/ASTP, OCR, and CMS creates compliance costs that large incumbents can absorb but smaller innovators cannot. Harmonizing requirements through a "OneHHS" approach and making guidance proportionate to risk would level the playing field for competition.

Barrier 1: Workforce Capacity Gap

Most healthcare delivery organizations (HDOs) lack the people, skills, and organizational processes needed to adopt AI responsibly. AI expertise remains concentrated in well-resourced academic medical centers, while community hospitals, federally qualified health centers (FQHCs), rural facilities, and safety-net providers often have no dedicated staff to evaluate, implement, or monitor AI tools. Without trained personnel and standardized processes, these organizations cannot determine whether an AI tool is safe, effective, and fair for their patient populations.

A nationwide analysis of 3,560 US hospitals found that nearly half reported no evaluation of AI model accuracy (47%) and over half reported no evaluation for bias (51%), with hospitals in medically underserved areas and health professional shortage areas significantly less likely to have adopted AI at all.¹ Among hospitals within HAIP's community of practice, our survey also found that only 61% conduct local evaluations for accuracy, and just 44% assess for fairness.² Interviews across 10 healthcare organizations revealed no consensus on how to evaluate whether AI performs fairly for all patients.³ The HEAAL framework, a structured fairness assessment protocol, found that rigorous evaluation requires investment "beyond reach for most organizations" without external support.⁴

HAIP has developed freely available best-practice guides covering the full AI lifecycle, organized around 8 key decision points.⁵ Through its Practice Network, HAIP operates a hub-and-spoke model in which experienced healthcare delivery organizations in the Corps Network serve as hubs, providing mentorship on governance, implementation, and workflow design, while organizations new to AI serve as spokes, receiving structured support to build local capacity. In its inaugural cohort, HAIP paired 5 safety-net providers with mentors from 15+ organizations, building foundational governance capacity, including AI use policies, governance committees, and staff training at each site.

There is an opportunity for HHS to support AI technical assistance programs built on public-private partnerships. Our experience shows that the hub-and-spoke model is effective for disseminating knowledge, governance best practices, and organizational readiness, but implementation, monitoring, and ongoing management require technical infrastructure and expertise that go beyond what hub organizations alone can provide. A sustainable model would pair hub-and-spoke knowledge networks with cloud and AI technology partners that provide compute infrastructure, technical tooling, and deployment expertise, while adopting institutions contribute their local clinical context, workflow requirements, and community-specific needs. Programs such as HRSA's Telehealth Resource Centers and ONC's Regional Extension Centers demonstrate that this approach works when each partner contributes its strengths.

¹Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

²Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity to address community health challenges." *PLOS Digital Health* (2025).

³Kim JY, et al. "Organizational governance of emerging technologies: AI adoption in healthcare." *ACM FAccT* (2023).

⁴Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

⁵Health AI Partnership Key Decision Points. <https://healthaipartnership.org/key-decisions-in-adopting-an-ai-solution>

Without capacity-building before compliance mandates, providers serving Medicare and Medicaid populations risk falling further behind, widening the access-to-care and quality gaps.

Barrier 2: Data Infrastructure and Interoperability

Even when organizations have the workforce to adopt AI, many lack the underlying data infrastructure. AI tools require access to structured, high-quality health data, but healthcare data remains fragmented across systems, locked behind proprietary platforms, and difficult to share. Smaller providers are particularly affected: over 20% of office-based practices still lack certified electronic health record (EHR) technology, the most basic foundation for AI.⁶

ONC data shows persistent gaps in health IT adoption among smaller practices and safety-net providers.⁷ The lack of data interoperability means that AI tools validated at one institution often cannot be reliably deployed at another without significant adaptation, a barrier that disproportionately burdens organizations with fewer technical resources. Through its Practice Network, HAIP has seen these infrastructure gaps up close, particularly at safety-net sites where data readiness is often the first barrier to AI adoption, well before questions of governance or clinical validation arise.

Continued advancement of data interoperability standards through ONC/ASTP, with specific attention to ensuring AI tools can access and operate on health data across different systems, would meaningfully reduce this barrier. There is also an opportunity to explore pathways for healthcare organizations to use protected health information on shared computing infrastructure in a cost-effective, HIPAA-compliant manner, recognizing that computational costs may be prohibitive for smaller providers operating on thin margins.

Barrier 3: Misaligned Payment Incentives

There is no clear mechanism to reimburse healthcare organizations for the work required to adopt AI responsibly. Fee-for-service payment does not reward AI efficiency gains, and there is no financial support for the substantial effort required to validate AI tools locally and assess their safety and fairness. This creates a perverse incentive: organizations that skip rigorous evaluation can deploy faster and at lower cost than those that do it properly.

Organizations that conduct thorough AI evaluation, including local validation, safety review, and fairness analysis, bear higher costs with no financial return under current payment models.⁸ Some experts have proposed value-based and outcome-linked reimbursement approaches, but these models remain immature.⁹ Meanwhile, mixed evidence on AI return on investment suggests that current payment

⁶Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity to address community health challenges." *PLOS Digital Health* (2025).

⁷Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity to address community health challenges." *PLOS Digital Health* (2025).

⁸Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

⁹Hasan A, et al. "Aligning AI principles and healthcare delivery organization best practices." *npj Digital Medicine* (2025).

structures do not yet capture the full value AI can deliver.¹⁰ HAIP is beginning to track implementation costs and outcomes across its Practice Network sites, with the goal of generating real-world evidence on the resources required for responsible AI deployment. This data does not yet exist at scale, which is itself part of the problem: without clear evidence on what responsible adoption costs, it is difficult to design payment models that account for it.

Under value-based care, the incentives align more naturally: better outcomes translate to financial reward, which makes the upfront investment in governance and validation worthwhile. Existing value-based programs like ACO and MSSP frameworks offer a natural vehicle for recognizing AI-driven outcome improvements without creating new standalone reimbursement mechanisms. But with fee-for-service still dominant across much of the healthcare system, most organizations have no business case for investing in AI adoption, let alone doing it responsibly. As the evidence base matures, HAIP's ongoing work to quantify implementation costs and outcomes across its Practice Network can help inform how these programs account for AI value.

Barrier 4: Regulatory Fragmentation

Healthcare organizations face fragmented AI guidance from multiple HHS agencies, each issuing separate requirements. External frameworks from NIST, WHO, and state attorneys general add further complexity. The resulting confusion is especially burdensome for smaller organizations that lack dedicated regulatory or legal staff to navigate overlapping, sometimes conflicting requirements.

HAIP's research mapped 31 best practices against 8 regulatory frameworks, distilling 58 raw principles down to 13 synthesized principles and translating them into step-by-step procedures with identified stakeholders and data sources, revealing substantial redundancy and inconsistency across frameworks.¹¹ State-level actions, including the California Attorney General's inquiry into AI bias and the Colorado AI Act, add yet another layer of complexity.¹² HAIP has maintained ONC as an ecosystem partner since its founding and has engaged legal and regulatory experts from DLA Piper throughout its framework development.

Advancing the "OneHHS" approach described in the HHS AI Strategy by harmonizing guidance across FDA, ONC/ASTP, OCR, and CMS into coordinated, actionable guidance would significantly reduce this burden. Translating high-level principles into practical, step-by-step implementation guidance would give organizations a clear path forward. Making compliance requirements proportionate to risk and pairing them with capacity-building support would help ensure that under-resourced providers are not penalized for lacking the resources to navigate regulatory complexity.

¹⁰Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity to address community health challenges." *PLOS Digital Health* (2025).

¹¹Hasan A, et al. "Aligning AI principles and healthcare delivery organization best practices." *npj Digital Medicine* (2025).

¹²Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity to address community health challenges." *PLOS Digital Health* (2025).

Question 2: Policy Changes

Summary

- **Regulatory harmonization:** A "OneHHS" approach that harmonizes requirements across FDA, ONC/ASTP, OCR, and CMS, preserves market transparency as certification is streamlined, enforces information-blocking rules to enable independent AI developers to compete on merit, and modernizes HIPAA to clarify permissible uses of data for AI development would level the playing field for innovators of all sizes.
- **Payment alignment:** Value-based programs offer a natural vehicle for incentivizing responsible AI adoption by tying payment to demonstrated outcomes rather than creating standalone reimbursement codes. The ACCESS Model's outcome-aligned payments point in this direction, but broader recognition of AI-enabled interventions across existing CMS programs is needed.
- **Capacity-building infrastructure:** A national hub-and-spoke network for AI technical assistance, embedded within existing federal programs such as Rural Health Transformation, would build capacity within under-resourced organizations without expanding government. Capacity-building needs to precede compliance mandates to avoid penalizing the organizations least equipped to comply.
- **Agentic AI and autonomous care:** AI in healthcare is moving from decision support toward autonomous clinical action, as illustrated by Utah's AI prescription renewal program and ARPA-H's ADVOCATE program. Current regulatory, payment, and governance frameworks are not designed for AI that acts independently. HHS will need to address how to classify and oversee autonomous clinical AI, how to structure liability, when human oversight is required and when it is not, and how to reimburse care delivered by AI agents rather than practitioners.

Regulatory Changes

As discussed in Question 1, healthcare organizations face fragmented AI guidance from multiple HHS agencies, each issuing separate requirements alongside external frameworks from NIST, WHO, and state attorneys general. Here we focus on specific regulatory changes that would reduce this burden while preserving the market conditions needed for competition and innovation.

The HTI-1 Final Rule (45 CFR § 170.315(b)(11)) took a meaningful step by requiring certified health IT developers to disclose source attributes for predictive decision support interventions, enabling users to assess whether tools are Fair, Appropriate, Valid, Effective, and Safe (the FAVES framework). However, these transparency requirements apply only to certified health IT, leaving a large and growing category of clinical AI tools outside their scope, including generative AI applications like ambient scribes that are deployed across hundreds of health systems without FDA approval and with minimal transparency requirements. The subsequent withdrawal of remaining HTI-2 proposals and publication of the HTI-5 Proposed Rule in December 2025 signal a welcome shift toward streamlining certification requirements and reducing compliance burden on health IT developers.¹³ However, HTI-5's proposal to eliminate source-attribute disclosure requirements for predictive decision-support interventions risks eroding the market transparency that enables competition.

HAIP's experience piloting the AI Vendor Disclosure Framework with safety-net hospitals and FQHCs illustrates why.¹⁴ Without structured disclosure, organizations with limited technical staff cannot independently evaluate vendor claims about model performance, training data, or fairness, and default to the AI bundled with their EHR, not because it is superior, but because evaluating alternatives requires expertise they lack. Standardized transparency gives these organizations and independent AI developers a common basis for comparing products on merit. Removing it does not reduce the burden on healthcare purchasers; it shifts the advantage toward incumbents. As discussed in our response to the capacity-building question below, building local capacity to evaluate AI tools is an essential complement to streamlined certification; without both, competitive markets will not be achieved.

The transparency gap is compounded by a lack of data access. Enforcement of information blocking rules (45 CFR Part 171) has not kept pace with the extent to which dominant EHR vendors control access to clinical data and workflow integration points. As we have previously documented, this creates a market failure: well-validated AI built outside EHR platforms struggles with adoption, while less rigorously tested AI bundled into EHR systems achieves rapid uptake simply because it is cheaper to deploy.¹⁵ A

¹³Health Data, Technology, and Interoperability: ASTP/ONC Deregulatory Actions to Unleash Prosperity (HTI-5 Proposed Rule), 90 Fed. Reg. 60970 (proposed Dec. 29, 2025).
<https://www.federalregister.gov/documents/2025/12/29/2025-23896>

¹⁴Kpodzro S, et al. "A Collaborative Best Practice Guide for Promoting AI Vendor Transparency in Healthcare: The HAIP AI Vendor Disclosure Framework." SSRN (2025).
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5440315

¹⁵Sendak MP, Balu S, Price WN II. "A market failure is preventing efficient diffusion of health care AI software." *STAT News* (2022). <https://www.statnews.com/2022/05/24/market-failure-preventing-efficient-diffusion-health-care-ai-software/>

recent state antitrust action against a major EHR vendor alleges exactly this pattern.¹⁶ Without stronger enforcement, small and mid-sized AI developers will remain unable to compete on the merits of their technology.

HIPAA's privacy framework, written well before AI, adds another layer of uncertainty. Healthcare organizations and AI developers face unclear rules about how protected health information can be used for model development, local validation, and ongoing monitoring, all activities essential to responsible AI adoption. Smaller developers and community providers, which cannot afford specialized legal counsel to navigate these ambiguities, are disproportionately affected. Modernizing HIPAA to clarify permissible uses of health data for AI development and validation while maintaining patient privacy protections would reduce regulatory uncertainty and enable broader participation in AI innovation.

A "OneHHS" approach to regulatory harmonization, as articulated in the December 2025 HHS AI Strategy, would address the most immediate barriers. Key priorities include: (1) as the HTI-5 rulemaking streamlines certification, preserving and extending source attribute transparency requirements to cover all clinical AI tools that substantively influence care decisions, building on the HTI-1 FAVES framework so that a modernized program retains the market transparency needed for fair competition; (2) strengthening enforcement of information blocking rules (45 CFR Part 171) to ensure that EHR vendors cannot use data control to disadvantage independent AI developers, leveling the playing field for market competition and innovation; and (3) translating high-level principles into practical, step-by-step guidance proportionate to risk, so that compliance is achievable for organizations of all sizes rather than only those with the resources to hire dedicated regulatory teams. Relevant regulations include 45 CFR Parts 164, 170, and 171, and 21 CFR Parts 800–898.

Payment Policy Changes

Clinical AI spans direct patient care (e.g., sepsis prediction, diagnostic imaging), operations (ambient scribes, care coordination), and administration (coding, prior authorization), and its value shows up differently in each: cost savings in some cases, provider well-being in others, improved quality in still others. Fee-for-service reimbursement was not designed for any of these, and there is no straightforward billing mechanism for an ambient scribe that reduces burnout or a care coordination tool that avoids downstream utilization. Many of AI's strongest applications, including prevention, risk stratification, care coordination, and early intervention, are systematically under-supported in fee-for-service payment, which rewards volume rather than the proactive, longitudinal care that AI enables.

Value-based and capitated payment models create stronger alignment with AI's strengths, but current programs do not yet explicitly recognize AI-enabled interventions as qualifying activities. Meanwhile, with shrinking margins, health systems rationally minimize adoption costs by skipping validation and monitoring altogether, or defaulting to whatever AI their EHR vendor bundles in. The result is widespread adoption of poorly governed AI, reinforcing the vendor lock-in and market concentration problems discussed in the regulatory section above.

¹⁶State of Texas v. Epic Systems Corporation, No. 236-372872-25 (Tarrant County District Court, filed Dec. 11, 2025).

Several payment shifts within existing CMS programs would incentivize responsible AI adoption: (1) adding MSSP quality measures that recognize validated AI use (42 CFR Part 425); (2) creating MIPS Improvement Activities crediting clinicians for implementing AI tools meeting HTI-1 transparency standards (42 CFR § 414.1380); (3) using CMMI authority (§ 1115A) to test payment models that reward demonstrated AI value, building on the ACCESS Model precedent; (4) within Medicare Advantage (42 CFR Part 422) and Medicaid managed care (42 CFR Part 438), explicitly allowing AI-enabled interventions, such as predictive outreach, risk identification, and care navigation, to count toward performance measures and medical loss ratio activities, so that organizations investing in these capabilities are not penalized under existing program rules; (5) providing time-limited downside risk protection for providers adopting AI-enabled care transformation, particularly safety-net providers serving high-need populations who lack the reserves to absorb losses during the potentially long runway before AI demonstrates durable value (CMS Innovation Center models under 42 CFR Part 512; HRSA cooperative agreements under 42 CFR Part 51c); and (6) revisiting narrow definitions of reimbursable services that exclude AI-enabled preventive interventions, such as predictive outreach, social risk identification, and early intervention, so that AI tools supporting prevention and cost containment are not structurally disadvantaged under current benefit definitions (42 CFR Part 410; 42 CFR §440.130).

Programmatic Design Changes

The most immediate barrier to AI adoption is capacity. Most healthcare organizations, particularly those in rural and underserved communities, lack the technical personnel, data infrastructure, and organizational processes to evaluate, implement, and govern AI tools. Community demand for AI is already emerging: patients and families are asking for AI-powered tools like multilingual virtual assistants that provide support and connect them to community resources, but most organizations lack the capacity to deploy such tools responsibly. Because AI implementation and health policy operate primarily at the state level, federal infrastructure for AI capacity-building, embedded within existing programs like Rural Health Transformation, would address this gap most effectively.

The Rural Health Transformation Program offers a natural venue for this work. Its digital modernization investments are already creating state-level structures, including technical assistance centers, regional hubs, and innovation funds, that require AI governance expertise to function effectively.¹⁷ HAIP and DIHI have direct experience building this kind of capacity. As described in Question 1, HAIP's Practice Network hub-and-spoke model has built foundational governance at safety-net sites, and its inaugural cohort launched AI tool pilots at 3 of 5 sites within 12 months.

Key elements of this infrastructure include: (1) a national hub-and-spoke network for AI technical assistance, with centers of excellence providing governance frameworks, evaluation standards, and mentorship to regional and state-level partners; (2) embedding AI readiness criteria within existing federal programs (Rural Health Transformation, Telehealth Resource Centers, AHRQ collaboratives) so that capacity-building is integrated into digital modernization investments rather than siloed; (3) funding research on AI applications that address community health needs, including patient-facing tools designed for the populations that community health organizations serve; and (4) building foundational AI capacity,

¹⁷North Carolina Department of Health and Human Services. "North Carolina Rural Health Transformation Program Application." (2025). <https://www.ncdhhs.gov/divisions/office-rural-health/rural-health-transformation-program>

including governance, workforce skills, and technical infrastructure, as a prerequisite for meaningful compliance rather than imposing requirements that under-resourced providers cannot meet.

Looking Ahead: Policy Frameworks for Autonomous and Agentic AI

The regulatory, payment, and capacity challenges discussed above assume AI systems that assist or augment human clinicians. But AI in healthcare is already moving toward autonomous clinical action. Utah's AI regulatory sandbox recently approved the first program allowing an AI system to independently renew prescription medications for chronic conditions, with no physician in the decision loop for routine renewals.¹⁸ ARPA-H's ADVOCATE program is developing agentic AI for cardiovascular care with a 39-month timeline to FDA authorization.¹⁹ Combined with the ACCESS Model's outcome-aligned payments and the FDA's TEMPO enforcement discretion pilot, these developments point toward a near-term future in which AI systems act as independent clinical agents, not just decision support tools. The policy framework is not keeping pace, and HHS will need to address this across all three dimensions: regulatory, payment, and programmatic.

On the regulatory side, a threshold question is how to classify autonomous clinical AI. The current FDA framework regulates AI as Software as a Medical Device (SaMD), which assumes a human clinician interpreting or acting on the AI's output. When the AI itself makes the clinical decision, that framework breaks down. The Healthy Technology Act (H.R. 238) takes the opposite approach, proposing that AI systems qualify as practitioners authorized to prescribe medications.²⁰ These are fundamentally different regulatory models with different implications for oversight, accountability, and market structure. Closely related is the question of human oversight. Not all clinical decisions carry the same risk, and a one-size-fits-all requirement for physician-in-the-loop would prevent AI from reaching the settings where it is most needed, particularly underserved communities with severe clinician shortages.

Liability is equally unresolved. The medical malpractice framework assumes a physician-patient relationship. When an autonomous AI system makes a clinical error, it is unclear whether liability falls on the developer, the deploying institution, the supervising physician, or some combination. Utah's approach places liability on the AI company (Doctronic carries malpractice insurance covering the AI system), but this is an ad hoc state-level solution. Without a federal framework, liability rules will develop inconsistently across states, creating the same kind of regulatory fragmentation that already burdens the healthcare AI market.

On payment, the existing reimbursement system is built around paying practitioners for services. CPT codes assume a human clinician performing or supervising work. The sole precedent for autonomous AI reimbursement is CPT 92229, created for an autonomous diabetic retinopathy detection system that

¹⁸Utah Department of Commerce, Office of AI Policy. "Utah and Doctronic Announce Groundbreaking Partnership for AI Prescription Medication Renewals." (Jan. 6, 2026). <https://commerce.utah.gov/2026/01/06/news-release-utah-and-doctronic-announce-groundbreaking-partnership-for-ai-prescription-medication-renewals/>

¹⁹ARPA-H. "Agentic AI-Enabled Cardiovascular Care Transformation (ADVOCATE)." (Jan. 2026). <https://arpa-h.gov/explore-funding/programs/advocate>

²⁰Healthy Technology Act, H.R. 238, 119th Congress (2025). <https://www.congress.gov/bill/119th-congress/house-bill/238>

operates without a physician at the point of service.²¹ CPT Appendix S now classifies AI services across a spectrum from assistive to augmentative to autonomous²², but the payment infrastructure for the autonomous end of that spectrum remains largely theoretical. As agentic AI enters clinical practice, CMS will need to determine how outcome-based payment models account for the quality of AI governance, not just downstream clinical results.

On the programmatic side, the governance infrastructure needed for agentic AI is substantially greater than what current AI tools require. The organizations that will be asked to deploy autonomous AI systems need governance capacity now, before agentic AI arrives at scale. Building that capacity, through the programmatic investments described above, is the most important thing HHS can do to ensure that autonomous AI is deployed responsibly when it is ready.

²¹Digital Diagnostics. CPT 92229 for autonomous AI diabetic retinopathy detection established the first nationally reimbursed autonomous AI clinical service through Medicare.

²²American Medical Association. "CPT Appendix S: AI Taxonomy for Medical Services and Procedures." Effective January 2022.

Question 3: Legal and Implementation Issues (Non-Medical Devices)

Summary

- **Liability framework misaligned with AI reality:** Standard vendor contracts shift risk entirely to providers through limitation-of-liability and hold-harmless provisions, creating an uneven playing field that disadvantages smaller innovators and under-resourced health systems. AI compounds the problem by making it harder to identify when and how harm occurs. Model contract language tied to CMS participation requirements, combined with AI incident reporting, would level the playing field without requiring new statutory authority.
- **Non-device AI lacks meaningful oversight:** The CDS exclusion framework was designed before generative AI and assumes clinicians actively verify AI outputs. The 2026 FDA CDS guidance made important advances, but ambiguity remains for AI that avoids the CDS label entirely or whose outputs cannot be meaningfully reviewed. Automation bias already undermines the premise that human-in-the-loop oversight is sufficient, and that premise breaks down entirely for agentic AI that acts without human intermediation. Classification based on clinical function, and actual clinical workflow impact rather than manufacturer labeling would give developers the predictability they need to invest and compete.
- **Governance structures not designed for AI:** Non-device AI falls between existing institutional governance structures, and every organization currently builds AI governance from scratch. This costly duplication slows adoption, burdens health systems that want to deploy AI responsibly, and creates unpredictable procurement environments for vendors. As autonomous AI enters clinical care, governance needs will only intensify. There is an opportunity for HHS to support the development of standardized governance models that reduce this burden on individual organizations while giving vendors more predictable markets.

Liability Framework Misaligned with AI Reality

The current liability landscape for clinical AI is uncertain and unevenly distributed. Vendor contracts routinely include limitation-of-liability and hold-harmless provisions that shift risk entirely to providers. Large incumbent vendors have the legal resources to insulate themselves, while smaller innovators and under-resourced health systems absorb liability by default. AI adds a further complication: when AI systems contribute to adverse outcomes, the complexity of these tools can make it difficult to determine that harm has occurred, who is responsible, and what went wrong.²³ This uncertainty discourages both developers and health systems from pursuing responsible AI adoption.

Because liability is primarily a matter of state law, HHS's direct role is limited. However, there are practical steps within existing authorities that would reduce uncertainty and level the playing field. Developing model contract language for AI procurement, tied to CMS conditions of participation or government procurement requirements, would help ensure fair risk allocation and particularly benefit smaller AI developers and community health systems that lack resources to negotiate bespoke agreements. Establishing AI incident reporting requirements would help illuminate when and how AI-related harms occur. Recognizing organizations that follow established AI governance frameworks through existing quality program incentives, such as the CMS Quality Payment Program, would encourage responsible adoption without requiring new statutory authority.

Looking further ahead, as AI takes on more autonomous clinical roles, a federal compensation program modeled on the National Vaccine Injury Compensation Program (VICP) could provide a mechanism for compensating patients harmed by AI while reducing liability uncertainty. This would require legislation, but it is within HHS's policy domain and worth advancing as AI's clinical role expands.

Non-Device AI Lacks Meaningful Oversight

The CDS exclusion framework was a sound deregulatory step, and FDA's updated CDS guidance (January 2026) made further progress by introducing a "meaningful independent review" standard and clarifying that generative AI outputs may fall outside the non-device exclusion.²⁴ However, regulatory ambiguity remains and creates real costs for innovation. Developers, particularly startups, cannot easily predict whether their products will be classified as devices, making it difficult to plan development timelines, raise capital, or bring products to market efficiently.

Some areas remain unclear. The enforcement discretion policy does not cover time-critical scenarios or AI systems whose basis cannot be meaningfully reviewed, precisely where generative AI raises the greatest concern. The framework still relies on manufacturers' stated "intended use" to determine classification, allowing companies to position clinically influential products outside the CDS framework entirely. AI systems that avoid the CDS label altogether while still influencing clinical decisions through workflow integration are not addressed. This ambiguity disadvantages responsible developers who seek clarity while allowing less scrupulous competitors to exploit gray areas.

²³Price WN. "Artificial Intelligence in Health Care: Applications and Legal Issues." *SciTech Lawyer* (2019).

²⁴FDA. "Clinical Decision Support Software: Guidance for Industry and Food and Drug Administration Staff." (January 6, 2026).

Building on the 2026 guidance, there is an opportunity to reduce this ambiguity by adopting a more function-based approach to classification, where actual clinical workflow impact rather than manufacturer labeling determines the level of oversight.

More fundamentally, the entire oversight architecture for non-device clinical AI rests on the premise that a clinician independently reviews AI outputs before they affect patient care. Automation bias already strains this premise, and for generative AI, where outputs cannot be traced to verifiable sources, meaningful independent review may be impossible even when clinicians attempt it. As AI moves toward agentic and autonomous clinical roles, as discussed in Question 2, there is no human review step to rely on at all. Human-in-the-loop oversight remains important, but it cannot serve as the sole regulatory strategy. Additional safeguards, including outcome-based monitoring, automated safety checks, and override capabilities, are needed to complement human oversight where it exists and substitute for it where it does not.

Governance Structures Not Designed for AI

Non-device AI falls between existing HDO governance structures in ways that create real barriers to responsible adoption. Medical device committees oversee FDA-regulated products but have no jurisdiction over non-device AI. IT governance manages technology procurement and security but typically lacks the clinical expertise to evaluate AI's impact on care decisions. Clinical practice committees set treatment protocols, but AI represents a novel technology category without established clinical standards. Compliance and risk management offices handle regulatory obligations, but there is no clear regulatory framework for non-device clinical AI to comply with. The result is that AI governance responsibility is fragmented across multiple internal committees and external stakeholders, including vendors, integrators, and clinicians, each holding partial jurisdiction with none having complete oversight or clear accountability.

This fragmentation means every HDO must create new governance structures from scratch, a costly duplication of effort with no standardized models to follow. HAIP's research across 10 HDOs found no consensus on how to govern AI, with organizations independently "figuring it out" through trial and error.²⁵ The development of HEAAL demonstrated the substantial investment in personnel, data infrastructure, and organizational capacity that rigorous AI governance demands, an investment that stretches the budgets of even well-resourced organizations.²⁶ Separate research on algorithm deployment found that regulatory affairs involvement is often ad hoc rather than systematic, brought in too late to shape implementation decisions.²⁷ The burden of this piecemeal approach falls hardest on under-resourced health systems, including rural hospitals, safety-net providers, and community health centers, which face the same governance challenges as large academic medical centers but with a fraction of the

²⁵Kim JY, et al. "Organizational governance of emerging technologies: AI adoption in healthcare." *ACM FAccT* (2023).

²⁶Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

²⁷Boag W, et al. "The Algorithm Journey Map." *npj Digital Medicine* (2024). <https://doi.org/10.1038/s41746-024-01061-4>

personnel and expertise. For AI vendors, particularly smaller companies and startups, the lack of standardized governance creates unpredictable procurement requirements that vary across every customer, raising the cost of market entry and disadvantaging innovators who cannot afford to navigate idiosyncratic institutional processes.

Most governance attention has also focused on pre-deployment evaluation, with far less attention to post-deployment workflow optimization and impact measurement. A predictive model that accurately identifies patients at risk of falling will not reduce falls if the clinical workflow does not ensure timely, effective intervention. Post-deployment governance, including workflow integration, outcome monitoring, and ongoing optimization, is where AI's potential benefit is most likely to be lost and where harm is most likely to occur.

There is an opportunity for HHS to reduce the governance burden on individual organizations by supporting the development of model AI governance structures that HDOs can adapt to their own context and scale. Rather than requiring every organization to build frameworks independently, standardized models would allow health systems to focus their limited resources on implementation rather than framework design. Standardized governance would also benefit AI vendors by creating more predictable procurement environments, lowering the cost of market entry and enabling smaller companies to compete on the quality of their technology rather than their ability to navigate idiosyncratic institutional processes. Guidance on AI documentation requirements for clinical settings, specifically what to record about AI's role in care decisions, would reduce liability exposure and improve post-deployment monitoring. These are the kinds of enabling conditions that encourage responsible AI adoption without expanding government mandates.

These governance frameworks also need to evolve for autonomous AI. Current frameworks, including HAIP's, were designed for AI that augments human decisions. As AI moves toward autonomous clinical roles, as discussed in Question 2, governance will need to address fundamentally new questions: how to establish accountability when there is no human in the decision loop, how to monitor and override autonomous systems, how to document AI-initiated clinical actions, and how to ensure that governance structures keep pace with rapidly advancing capabilities. The governance capacity that HAIP's Key Decision Points framework and Practice Network represent becomes more critical, not less, as AI autonomy increases. HHS can accelerate this work by supporting the development of next-generation governance frameworks through the same mechanisms described throughout this response: technical assistance networks, model governance structures, and capacity-building at under-resourced organizations. As emphasized throughout, capacity-building needs to come before compliance mandates.

Question 4: AI Evaluation Methods (Non-Medical Devices)

Summary

- **Pre-deployment local validation:** Vendor-reported performance metrics may not reflect how an AI tool performs within a given organization's patient population and clinical context, yet a national analysis of 3,560 hospitals found that fewer than a third evaluate AI accuracy across all their models. Grants to build validation infrastructure and cooperative agreements to enable shared learning across organizations would address this capacity gap most directly.
- **Fairness metrics and assessment:** AI models can produce meaningfully different predictions across patient subgroups, yet only 22% of U.S. hospitals assess for these performance discrepancies across all their AI models. No consensus exists on which metrics to use, because the appropriate measures depend on each organization's patient population and the AI use case. Investment in demographic data infrastructure and cooperative agreements for multi-site evaluation would enable organizations to move from working in isolation to learning from each other.
- **Post-deployment monitoring:** AI performance drifts as patient populations and clinical practices change, yet a national analysis found that only about a quarter of hospitals monitor all their AI models after deployment, and nearly half report no monitoring at all. Inadequate monitoring is a key reason many AI implementations remain stuck in the pilot phase. Cooperative agreements for shared monitoring infrastructure and grants for longitudinal outcome studies would help organizations move beyond ad hoc approaches.
- **Human-centered evaluation:** AI tools are sociotechnical systems whose impact depends on workflow integration, user trust, and organizational context. As AI moves toward greater autonomy, evaluation methods also need to address scenarios where there is no human reviewer. Grants for human factors research and technical assistance for under-resourced organizations would strengthen the evidence base for effective AI integration.

Pre-Deployment Local Validation

Vendor-reported performance metrics may not reflect the value and risks an AI solution presents within a given organization's own patient population and clinical context. Each organization needs to assess these locally, yet most lack the capacity to do so rigorously. National survey data shows fewer than a third of hospitals evaluate AI accuracy across all deployed models, and nearly half have no accuracy evaluation process at all.²⁸ Rigorous evaluation requires significant investment in personnel, capabilities, and data infrastructure that is beyond reach for most organizations, even well-resourced ones.^{29,30} No standardized procedures exist for what local validation entails. Some vendors have begun providing open-source validation tools, such as Epic's Seismometer, which HAIP sites have tested.³¹ However, effective use of these tools still requires underlying technical expertise and dedicated teams to implement, customize, and interpret results accurately. The lack of standardization and capacity underscores the need for shared networks where organizations serving similar patient populations can pool knowledge and resources for validation.

HAIP's HEAAL framework offers step-by-step guidance for evaluating AI tools across five domains: Reliability and Validity, Fairness, Fitness for Purpose, Transparency, and Accountability. With 37 procedures for existing AI and 34 for new AI development, the framework provides the kind of structured, practical evaluation protocol that individual organizations otherwise have to create from scratch, and it has been adopted by organizations across the health system.³² HAIP's best practice guides provide structured evaluation protocols with identified stakeholders and data sources for each stage of the AI lifecycle. These tools can serve as the basis for standardized local validation protocols, and aggregating best practices and use cases across organizations can help inform these standards over time.

Grants to build technical and data infrastructure, train the workforce, and support dedicated evaluation staff would directly address the capacity barrier. Cooperative agreements that enable organizations to share lessons learned and validation approaches would strengthen system-wide capacity, allowing under-resourced organizations to benefit from the experience of those further along. This is the kind of investment that accelerates responsible adoption by making evaluation feasible rather than mandating it before organizations have the means to comply.

Fairness Metrics and Assessment

The core fairness challenge in clinical AI is performance discrepancy: AI models can produce meaningfully different predictions, recommendations, or outputs across patient subgroups defined by race, ethnicity, age, sex, insurance status, or language preference. HEAAL's Fairness domain provides

²⁸Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

²⁹Gulamali FF, et al. "Eliminating the AI Digital Divide by Building Local Capacity." *PLOS Digital Health* (2025).

³⁰Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL)." *PLOS Digital Health* (2024).

³¹Epic Systems. "Seismometer: AI model evaluation with a focus on healthcare." GitHub (2024).
<https://github.com/epic-open-source/seismometer>

³²Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

procedures for detecting and measuring these discrepancies through subgroup analysis protocols and disparity impact assessments that evaluate whether AI will widen or narrow existing disparities. However, even within the framework, identifying appropriate fairness metrics is something each organization must determine based on its own patient population and the specific AI use case. There is no universal set of fairness metrics that can be prescribed in advance because the right measures depend on context, and this was one of the most frequent questions HAIP received from organizations adopting the framework.³³

The scale of this challenge is significant. Only 22% of U.S. hospitals assess for bias across all their AI models, and more than half report no bias assessment at all.³⁴ HAIP's research found no consensus across organizations on how to measure or address performance discrepancies, with each developing its own institutional practices independently.³⁵ Detecting these discrepancies requires granular demographic data that many organizations lack, often due to poor documentation workflows or barriers to accessing existing records. Open-source tools like Aequitas and Fairlearn exist but still require domain expertise and carefully selected cohorts to produce meaningful results. The result is that organizations are working in silos on one of the hardest problems in AI governance.

Investment in fairness assessment infrastructure through grants would help address the underlying capacity constraints: workforce development for staff with technical and ethics expertise to conduct assessments, infrastructure for demographic data collection and subgroup analysis, and resources for community engagement to understand patient needs and context. Cooperative agreements would enable organizations to share what they are learning, since pooling knowledge across sites working on the same hard questions is more efficient than each organization solving them alone. Multi-site fairness evaluation, where the same AI tool is assessed across diverse populations served by different organizations, would generate direct evidence on where and how performance discrepancies emerge.

Post-Deployment Monitoring

AI performance drifts as patient populations, clinical practices, and EHR systems change, making continuous monitoring essential to ensure safety, effectiveness, and fairness after deployment. The same national analysis found that only about a quarter of hospitals conduct post-deployment monitoring across all their AI models, and 47% report no monitoring at all.³⁶ Inadequate monitoring is a key reason many AI implementations remain stuck in the pilot phase, unable to advance from promising pilots to sustained, production-quality deployment. Organizations that cannot systematically track how AI tools perform over time cannot identify when performance degrades, when patient populations shift, or when a tool is no longer safe to use.

³³Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

³⁴Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

³⁵Kim JY, et al. "Organizational Governance of Emerging Technologies: AI Adoption in Healthcare." *ACM FAccT* (2023).

³⁶Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

Promising approaches include dashboards that track both AI performance and underlying data quality over time, statistical methods to detect performance drift, clinician feedback loops for reporting errors or concerns, and clearly defined decommissioning criteria. Investing in data quality monitoring and longitudinal trend analysis can help organizations identify data drift early, before it manifests as AI risk. The FDA's Predetermined Change Control Plans (PCCP) approach applies only to regulated devices, leaving non-device AI without comparable monitoring expectations. Organizations lack clarity on who is accountable for post-deployment monitoring, how to set thresholds for updating or decommissioning tools, and how to assess impacts on patient outcomes and clinical workflow.³⁷ Generative AI and emerging agentic AI systems present additional monitoring challenges: their outputs are less predictable, harder to benchmark against fixed performance standards, and can shift with model updates that occur outside the organization's control. Standardized approaches for monitoring these newer AI categories do not yet exist, and developing them is an area where federal investment could have significant impact.

The resource demands of ongoing monitoring compound the problem. Many organizations expend most of their resources on initial deployment, leaving limited capacity for sustained oversight. Under-resourced organizations often remain stuck in the pilot phase, unable to advance to full deployment. Even well-resourced organizations frequently rely on vendors for monitoring support. While vendor support is valuable, organizations need the capacity to evaluate vendor services, determine what to request, and retain ownership of oversight rather than outsourcing accountability.

Cooperative agreements that enable organizations to share monitoring approaches and establish benchmarks would help build systematic capacity across the field. Grants for longitudinal evaluation studies would generate evidence on how AI performance changes over time in real-world settings and its impact on patient outcomes and clinical workflow. Supporting the development of standardized monitoring frameworks for generative and agentic AI would give organizations practical tools rather than leaving each to develop its own approach.

Human-Centered Evaluation

AI tools are sociotechnical systems whose performance depends not only on technical accuracy but on how they are integrated into clinical workflows, aligned with organizational structures, and designed around the people who use and are affected by them. HAIP's Algorithm Journey Map documented the full sociotechnical process of AI implementation and found that workflow integration is a critical factor in whether AI succeeds or fails, and that engaging end users from the very beginning of the adoption journey is essential for identifying affected stakeholders and anticipating implementation challenges.³⁸

Automation bias leads clinicians to over-rely on AI recommendations. Alert fatigue, cognitive load, and trust calibration all influence whether AI improves or harms care, and AI may disrupt clinical workflows even when it is technically accurate.

As AI moves toward greater autonomy, human-centered evaluation also needs to evolve beyond the assumption that a human reviewer is always in the loop. For fully automated AI systems, evaluation must

³⁷Kim JY, et al. "Organizational Governance of Emerging Technologies: AI Adoption in Healthcare." *ACM FAccT* (2023).

³⁸Boag W, et al. "The Algorithm Journey Map." *npj Digital Medicine* (2024).

address how to monitor performance and safety without relying on clinician review as the primary safeguard. This includes automated quality checks, outcome tracking, override mechanisms, and protocols for escalation when systems operate outside expected parameters. Under-resourced organizations in particular need technical assistance to build evaluation capacity for these emerging AI categories, not just the current generation of decision-support tools.

Grants for human factors research would strengthen the evidence base on critical questions including how workflow integration affects adoption rates and clinical outcomes, how clinicians perceive and interact with different AI use cases, the impact of AI adoption on clinician productivity and patient care, strategies for mitigating automation bias, and best practices for training and change management. Cooperative agreements could support multi-site workflow studies that identify what works across different organizational contexts and patient populations.

Building Evaluation Capacity at Scale

The evaluation challenges described above share a common thread: individual organizations, especially under-resourced ones including FQHCs, community hospitals, and safety-net providers, cannot build comprehensive evaluation capacity alone. Through the Practice Network, HAIP is currently implementing HEAAL-based evaluation at 5 under-resourced sites (4 FQHCs, 1 community hospital), generating real-world evidence on the feasibility of rigorous AI evaluation in resource-constrained settings. The SepsisWatch case study at Duke documents the end-to-end process of integrating a deep learning tool into routine clinical care, from problem assessment through implementation and evaluation planning, illustrating what sustained AI lifecycle management looks like in practice.³⁹ These tools and experience could serve as the foundation for a national evaluation network where institutions with established expertise support under-resourced organizations through cooperative agreements, while grants fund local capacity at spoke sites.

HAIP has already demonstrated the viability of this kind of cross-sector collaboration. The consortium's industry partners, including Roche, have contributed to developing evaluation frameworks and co-investment approaches, and HAIP has worked with Epic on testing open-source validation tools at Practice Network sites. An industry co-investment model that brings HDOs, HHS, and private-sector partners together could leverage the resources and expertise of all parties to accelerate evaluation infrastructure while reducing the burden on any single one. There is also an opportunity to explore AI sandbox environments where developers and organizations that agree to share pre- and post-deployment performance data, including subgroup analyses and sociotechnical evaluation of impact on clinical teams, receive streamlined regulatory treatment. This kind of approach would incentivize transparency and evidence generation while giving developers and health systems a clear, predictable path to responsible deployment.

³⁹Sendak MP, et al. "Real-World Integration of a Sepsis Deep Learning Technology Into Routine Clinical Care: Implementation Study." *JMIR Medical Informatics* (2020).

Question 5: Supporting Private Sector Activities

Summary

- **Multi-stakeholder collaboratives:** Private sector collaboratives that bring together healthcare delivery organizations (HDOs), technology vendors, academics, and community representatives can develop shared AI governance resources more responsively than government programs working alone. Because solutions are driven by end users addressing real operational challenges, the resulting frameworks carry practical relevance and broad adoption. Grants and cooperative agreements to fund this collaborative infrastructure, with safety-net and rural health representation built in, would scale what is already working.
- **Vendor transparency and disclosure standards:** Information asymmetry between AI vendors and HDOs remains a barrier to responsible adoption and healthy market competition. Standardized vendor disclosure frameworks level the playing field for under-resourced organizations while giving vendors, particularly smaller innovators, clear and predictable expectations. Aligning federal requirements with industry-developed standards would extend their reach without creating duplicative systems.
- **Technical assistance programs:** Under-resourced organizations need hands-on support to build AI governance capacity, not just frameworks to follow. Hub-and-spoke technical assistance programs where institutions with established expertise support organizations earlier in their AI journey can adapt to local needs and evolve alongside the field. Federal funding through grants and cooperative agreements could enable nationally what philanthropy has demonstrated at modest scale.
- **Certification and accreditation:** Voluntary, industry-developed certification and credentialing programs create positive incentives for responsible AI governance without imposing regulatory burden. These programs can respond to organizational needs faster than federal regulation and build practitioner buy-in because standards come from the field. Recognizing private sector certifications in federal programs and linking them to meaningful incentives would drive adoption, particularly among organizations that cannot pursue certifications without a clear return.

Multi-Stakeholder Collaboratives

Private sector collaboratives can convene diverse stakeholders around shared AI governance goals more responsively than government programs working alone. By bringing together HDOs, technology vendors, academics, legal experts, safety-net providers, rural health systems, minority-serving institutions, and community representatives, these collaboratives develop shared resources that reduce duplicative effort across organizations. Because participation is voluntary and solutions are driven by end users addressing real operational challenges rather than imposed from the top down, the resulting frameworks carry greater practical relevance and broader adoption than prescriptive mandates.⁴⁰ Organizations across the healthcare system are independently developing their own AI governance practices, with each solving the same problems from scratch, a costly duplication that collaborative infrastructure is designed to prevent.⁴¹

A reasonable concern is whether organizations will share challenges and solutions or view one another as competitors. HAIP's experience demonstrates that structured collaboration can overcome this barrier when the collaborative is designed around shared problems rather than competitive interests. The consortium operates as a multi-stakeholder collaborative of HDOs alongside ecosystem partners including the American Medical Association, DLA Piper, and UC Berkeley, with ASTP/ONC serving as a federal observer.⁴² HAIP also collaborates with industry partners including Roche. This structure has produced freely available governance resources spanning the full AI lifecycle, from procurement through monitoring and decommissioning, including a fairness evaluation framework and the AI Vendor Disclosure Framework, demonstrating that collaboration designed around shared governance challenges can achieve broad impact without government direction.^{43,44}

Grants and cooperative agreements to fund collaborative infrastructure would directly support the operational costs of convening and coordinating multi-stakeholder efforts. Ensuring diverse representation, including safety-net providers, rural health systems, and community-based organizations, is essential because these settings collectively serve more patients than any individual health system and are often left out of AI design and evaluation. Federal agencies participating as observers, as ASTP/ONC already does with HAIP, can stay informed and ensure alignment with federal priorities without directing outcomes. Recognizing and amplifying outputs from successful collaboratives would lend federal visibility to frameworks that have demonstrated real-world effectiveness.

⁴⁰Sendak MP, et al. "Empowering US healthcare delivery organizations to safely, effectively, and responsibly use health AI through communities of practice." *PLOS Digital Health* (2024).

⁴¹Kim JY, et al. "Organizational Governance of Emerging Technologies: AI Adoption in Healthcare." *ACM FAccT* (2023).

⁴²Sendak MP, et al. "Empowering US healthcare delivery organizations to safely, effectively, and responsibly use health AI through communities of practice." *PLOS Digital Health* (2024).

⁴³Hasan A, et al. "Aligning AI principles and healthcare delivery organization best practices." *npj Digital Medicine* (2025).

⁴⁴Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

Vendor Transparency and Disclosure Standards

Information asymmetry between AI vendors and HDOs remains a significant barrier to responsible adoption and healthy market competition. National data reveals that most hospitals lack the capacity to evaluate vendor AI claims, with fewer than a quarter assessing AI fairness across patient populations.⁴⁵ Vendors, meanwhile, face unpredictable, ad hoc disclosure requests from each prospective customer. Standardized disclosure frameworks address both problems: they level the playing field so that under-resourced organizations can apply the same evaluation criteria as academic medical centers, and they give vendors, particularly smaller innovators who may lack the brand recognition of larger incumbents, clear and predictable expectations that reward product quality over marketing resources.

HAIP developed the AI Vendor Disclosure Framework, a community-informed procurement tool that structures vendor evaluation across key dimensions of system performance, data governance, and integration requirements. DIHI researchers created the Model Facts Label in 2020, the first AI product label in healthcare, which was cited in the HTI-1 Final Rule. HTI-1 requires disclosure of source attributes for predictive decision support interventions, a requirement that HAIP helped shape through its ongoing engagement with ASTP/ONC. The proposed HTI-5 rule would further refine these transparency requirements, and continued alignment between federal rules and industry-developed standards will be important as the regulatory landscape evolves.

Aligning federal requirements with industry-developed standards rather than creating parallel systems would reduce regulatory duplication and extend the reach of frameworks that have already demonstrated effectiveness. For example, referencing private sector disclosure frameworks in ASTP/ONC guidance or CMS conditions of participation would encourage adoption through existing federal programs without imposing additional regulatory burden. There is also a need for technical assistance that helps organizations, particularly smaller ones, understand and apply these disclosure requirements, since a standard is only useful if organizations have the capacity to use it. More broadly, ensuring that third-party AI vendors have fair access to health IT integration interfaces, without overly restrictive contractual obligations from incumbent EHR firms, would support the kind of competitive market that disclosure standards are designed to enable.

Technical Assistance Programs

Private sector technical assistance programs can deliver hands-on AI governance support in ways that adapt to local needs and evolve alongside the field. Hub-and-spoke approaches leverage existing expertise at institutions with established AI programs to support organizations earlier in their AI journey, while peer learning communities with shared curricula and mentorship build relationships and knowledge that sustain beyond program timelines.⁴⁶ Shared infrastructure for common functions like vendor evaluation and performance monitoring can reduce fragmentation across healthcare organizations and allow under-resourced settings to focus their limited capacity on governance and workflow integration rather than duplicating technical work that could be done collectively.

⁴⁵Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

⁴⁶Sendak MP, et al. "Empowering US healthcare delivery organizations to safely, effectively, and responsibly use health AI through communities of practice." *PLOS Digital Health* (2024).

HAIP's Practice Network demonstrates this approach. The program supports 4 federally qualified health centers (FQHCs) and 1 community hospital through a 12-month engagement, with hub institutions including Duke, Mayo Clinic, and Kaiser Permanente providing structured guidance on governance, best practices, and workflow design. One important lesson from the Practice Network is the value of offloading vendor evaluation to hub institutions and the collaborative itself, allowing health centers to focus on building governance capacity and monitoring practices rather than conducting technical assessments they are not resourced to perform. The experience of HAIP's FQHC partners also illustrates how collaborative development between health centers and early-stage AI startups can produce solutions tailored to safety-net settings in ways that differ fundamentally from traditional vendor-purchaser relationships. Currently funded by philanthropy, the Practice Network shows what is possible at modest scale and what federal funding could enable nationally.

Grants and cooperative agreements directed toward private sector technical assistance programs that reach safety-net providers, FQHCs, and rural facilities would maximize impact while keeping delivery in the hands of practitioners who understand local needs. Setting explicit goals for resource sharing and collaboration across healthcare organizations, particularly in primary care and rural areas, would further reduce fragmentation. Infrastructure investments through programs like the Rural Health Transformation Program can extend AI capabilities to community and rural settings, though it will be important that such programs benefit new market entrants and do not only advantage incumbent firms already experienced at navigating government contracts. Successful precedents exist: programs like HRSA's Telehealth Resource Centers and the Regional Extension Centers for EHR adoption demonstrate that public-private partnerships in healthcare technology can achieve national scale when properly supported.

Certification and Accreditation

Voluntary certification and accreditation programs can raise the bar for AI governance without imposing additional regulatory burden. Industry-developed certification for HDOs demonstrating rigorous AI governance, accreditation standards for AI evaluation processes, and credentialing, including stackable credentials, for workforce members with AI governance competencies all create positive incentives for responsible adoption. Because these standards are developed by practitioners, they carry greater buy-in than top-down requirements and can respond faster to organizational needs as the field matures.⁴⁷

A concrete model for what institutional AI certification could look like already exists in a familiar regulatory framework. HAIP researchers have proposed adapting the Clinical Laboratory Improvement Amendments (CLIA) structure for clinical AI, arguing that AI systems and laboratory tests share a fundamental characteristic: both are highly sensitive to local factors and require ongoing quality management by trained professionals.⁴⁸ Under this approach, healthcare organizations would hold tiered AI certification based on their capabilities, with requirements for local validation, proficiency testing across sites running the same AI tools, defined recalibration triggers, and regular quality monitoring. Organizations lacking internal AI expertise could reference a certified "AI Ops" unit at another

⁴⁷Sendak MP, et al. "Empowering US healthcare delivery organizations to safely, effectively, and responsibly use health AI through communities of practice." *PLOS Digital Health* (2024).

⁴⁸Jackson BR, et al. "Regulation of artificial intelligence in healthcare: CLIA as a model." *JAMIA* (2024).

institution, analogous to how smaller labs reference high-complexity laboratories. Existing accreditation bodies such as the Joint Commission could develop AI-specific inspection standards rather than creating new regulatory infrastructure. This distributed, institution-based model complements centralized FDA product review by placing quality management where data originates and where AI outputs affect patient care.

HAIP's HEAAL framework, with its 37 procedures for evaluating existing AI across five domains including fairness, reliability, and transparency, could serve as the basis for the evaluation standards within such a certification structure.⁴⁹ HAIP's best practice guides define what responsible governance looks like in practice across 8 key decision points, providing a foundation for credentialing standards.⁵⁰ HAIP is exploring the development of a formal certification program in partnership with accrediting bodies. An important design consideration is distinguishing between organizational accreditation, which evaluates whether an HDO has appropriate governance structures and processes, and workforce credentialing, which evaluates whether individual staff members have the competencies needed to fulfill AI governance roles. Both are necessary and serve different purposes.

Recognizing private sector certifications in federal programs would give organizations credit for voluntary standards they have already achieved. Linking certifications to meaningful incentives, such as favorable treatment in quality programs or streamlined regulatory pathways, would drive adoption, particularly among FQHCs and safety-net providers that cannot afford to pursue certifications without clear return on investment. Federal recognition can set expectations for the quality and rigor that certifications must demonstrate while preserving the flexibility that makes private sector standards responsive to the field's rapid evolution.

Scaling Private Sector Approaches with Federal Support

Across each of these areas, the most impactful federal role is not to build new programs but to fund successful private sector approaches, participate as a convener and connector, recognize private sector standards in federal programs, and scale proven approaches to reach more organizations. HAIP has built functioning infrastructure for each activity described above: a multi-stakeholder collaborative with diverse partners and federal engagement, a vendor disclosure framework now influencing federal policy, a technical assistance program reaching safety-net providers, and foundational work toward certification based on its evaluation frameworks and best practice guides. With support through grants and cooperative agreements, these approaches could scale nationally, expanding the Practice Network to more communities, developing formal certification in partnership with accrediting bodies, and broadening industry engagement to bring more vendors into developing shared standards. An industry co-investment approach that brings HDOs, federal agencies, and private-sector partners together could leverage the resources and expertise of all parties while reducing the burden on any single one.

⁴⁹Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

⁵⁰Hasan A, et al. "Aligning AI principles and healthcare delivery organization best practices." *npj Digital Medicine* (2025).

Question 6: AI Performance and Promising Tools

Summary

- **Evidence has not kept pace with adoption:** Over 1,250 FDA-authorized AI devices are on the market and the ambient AI scribe category alone generated \$600 million in 2025, yet rigorous evidence on clinical outcomes and cost-effectiveness remains sparse. Most research evaluates algorithm accuracy rather than clinical impact, making it difficult for organizations to make informed purchasing decisions and for genuinely superior products to differentiate themselves in a competitive market.
- **Operational AI is showing the clearest early returns:** AI tools for revenue cycle management, prior authorization, and patient access are demonstrating consistent efficiency gains. FQHCs have found AI-powered call centers particularly valuable given staffing shortages. Ambient AI scribes show strong evidence of burnout reduction, though business models increasingly center on revenue capture, and direct clinical-facing tools continue to emerge with uncertain return on investment.
- **Fairness, generalizability, and workforce concerns:** Nearly half of hospitals report no AI accuracy evaluation, and tools validated at academic medical centers may not perform equivalently elsewhere. Separately, significant efficiency gains from administrative AI have led some organizations to consider workforce reductions, raising questions about labor market impact in communities where healthcare organizations are a primary employer.
- **Implementation capacity is the core barrier:** The most significant challenge is not algorithm development but organizational capacity for implementation, governance, and monitoring. Under-resourced organizations lack dedicated personnel, data infrastructure, connectivity, and governance processes. Shared evaluation infrastructure, technical assistance, and attention to foundational AI readiness would address this more efficiently than expecting each organization to build capacity independently.

The Evidence Gap

The private sector has moved fast on clinical AI. Over 1,250 FDA-authorized AI/ML-enabled medical devices are on the market as of mid-2025,⁵¹ and the ambient AI scribe category alone generated \$600 million in 2025 with 2.4x year-over-year growth.⁵² Yet rigorous evidence on whether these tools actually improve patient outcomes or reduce costs has not kept pace. Most published AI research evaluates algorithm accuracy rather than measuring what matters to health systems, clinicians, and patients: clinical outcomes, workflow impact, and return on investment.⁵³ Randomized controlled trials remain limited, and most evidence is retrospective, drawn from single academic medical center sites.⁵⁴ A national analysis of 3,560 U.S. hospitals found that nearly half reported no evaluation of AI model accuracy and more than half reported no assessment for bias, pointing to widespread gaps in the evidence organizations need to make informed decisions.⁵⁵

This evidence gap creates a market problem. Without reliable performance data, healthcare delivery organizations (HDOs) cannot make informed purchasing decisions, and vendors developing genuinely superior products struggle to differentiate themselves from competitors making unsubstantiated claims. The organizations most affected are those with the fewest resources to conduct their own evaluations, which means the market's information asymmetry falls hardest on safety-net providers and community hospitals.

Where Clinical AI Is Delivering Value

AI tools deployed in clinical care extend well beyond direct patient-facing applications. Administrative and operational AI, particularly tools for revenue cycle management, prior authorization, claims processing, and patient access, is showing some of the most consistent early returns. These use cases carry lower clinical risk, lend themselves to clearer performance metrics, and address pressing workforce constraints. FQHCs in HAIP's Practice Network have reported that AI-powered call center tools are particularly valuable given chronic staffing shortages, allowing smaller organizations to maintain patient access at levels that would otherwise require additional hires they cannot afford. The efficiency gains in these operational areas are real, though they raise a separate set of questions about workforce impact discussed below.

Among clinical applications, ambient AI scribes represent healthcare AI's first breakout success, and the evidence base is now substantial. A three-arm randomized controlled trial of 238 physicians across 14 specialties found approximately 7% improvement in burnout scores, reduced cognitive workload, and a 9.5% decrease in time-in-note for the AI-assisted group compared to control.⁵⁶ A quality improvement study across 6 health systems with 263 providers showed burnout decreasing from 51.9% to 38.8%, with

⁵¹FDA. "Artificial Intelligence-Enabled Medical Devices." <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-enabled-medical-devices>

⁵²Menlo Ventures. "2025: The State of AI in Healthcare." (2025).

⁵³Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity." *PLOS Digital Health* (2025).

⁵⁴Hassan M, et al. "Barriers to and facilitators of artificial intelligence adoption in health care: Scoping review." *JMIR Human Factors* (2024).

⁵⁵Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

⁵⁶Lukac PJ, et al. "Ambient AI scribes in clinical practice: A randomized trial." *NEJM AI* (2025).

improved cognitive task load and reduced after-hours documentation time.⁵⁷ A separate RCT at UW Health found that ambient AI reduced after-hours work by approximately 30 minutes per day and improved clinician well-being on the Stanford Professional Fulfillment Index.⁵⁸ These results matter not only for clinician quality of life but for workforce retention at a time when healthcare faces persistent staffing shortages.

However, the cost picture is more complicated. The PHTI AI Taskforce concluded that the evidence base for ambient scribes improving efficiency through reduced documentation time "is limited" and that the financial impact on health systems remains "unclear."⁵⁹ Business models in this space are increasingly centered on revenue capture through more intensive coding rather than cost reduction, and payers have begun responding with automatic downcoding of evaluation and management claims, raising questions about whether the current value proposition is sustainable.⁶⁰ Safety also warrants attention: RCTs report that AI-generated notes occasionally contain clinically significant inaccuracies, most commonly omissions and pronoun errors, with one mild patient safety event reported.⁶¹

Medical imaging AI represents the largest category of FDA-cleared AI devices, yet real-world results have been mixed. A 2024 survey of leading U.S. health systems found that while most organizations have deployed AI in imaging and radiology, only 19% report a "high degree of success."⁶² More broadly, direct clinical-facing AI tools are increasingly entering the market across specialties, but the evidence base for their return on investment remains thin compared to operational applications.

Where Clinical AI Has Fallen Short

The most persistent shortcoming is the gap between algorithm performance and clinical impact. High accuracy in controlled research settings often does not translate to real-world benefit because the critical challenge is integrating AI into clinical workflows, not building a better model. Alert-based clinical decision support systems illustrate the problem: override rates routinely exceed 90% due to alert fatigue, rendering even technically sound tools clinically irrelevant.⁶³

Fairness remains a serious concern. Well-documented cases of algorithmic bias include racial bias in widely used risk prediction algorithms that systematically underestimated the health needs of Black patients.⁶⁴ Yet more than half of hospitals report no bias assessment for their AI models, and only 12% of

⁵⁷Olson KD, et al. "Use of ambient AI scribes to reduce administrative burden and professional burnout." *JAMA Network Open* (2025).

⁵⁸Afshar M, et al. "A Pragmatic Randomized Controlled Trial of Ambient Artificial Intelligence to Improve Health Practitioner Well-Being." *NEJM AI* (2025). <https://doi.org/10.1056/AIoa2500945>

⁵⁹Peterson Health Technology Institute. "Adoption of Artificial Intelligence in Healthcare Delivery Systems." (2025).

⁶⁰Adler-Milstein J, Aggarwal N. "Ambient AI scribes and the coding arms race." *npj Digital Medicine* (2025).

⁶¹Lukac PJ, et al. "Ambient AI scribes in clinical practice: A randomized trial." *NEJM AI* (2025).

⁶²Poon EG, et al. "Adoption of artificial intelligence in healthcare: Survey of health system priorities, successes, and challenges." *JAMIA* (2025).

⁶³van der Sijs H, et al. "Overriding of drug safety alerts in computerized physician order entry." *JAMIA* (2006).

⁶⁴Obermeyer Z, et al. "Dissecting racial bias in an algorithm used to manage the health of populations." *Science* (2019).

implementation studies even note risks of exacerbating existing disparities.⁶⁵⁶⁶ Without systematic fairness evaluation, AI tools risk widening the very gaps they could help close.

Generalizability across settings is a related failure point. AI validated at large academic medical centers may not perform equivalently at community hospitals, FQHCs, or rural facilities, yet only 61% of hospitals conduct local validation before deployment.⁶⁷ The problem is compounded for safety-net providers, many of whom do not manage their own EHR instance or can only afford less sophisticated systems, limiting which AI tools they can even run. This infrastructure gap means the organizations serving the most complex patient populations are often the least equipped to evaluate or deploy AI.

A newer concern is emerging around workforce impact. Where AI tools have delivered significant efficiency gains, particularly in administrative and operational functions, some organizations have begun considering workforce reductions. In rural and safety-net settings, where healthcare delivery organizations are frequently the largest local employer, these decisions carry consequences that extend well beyond the organization itself. AI-powered call centers, revenue cycle automation, and coding assistance can reduce costs, but if those savings come primarily through job displacement in economically fragile communities, the net benefit is less clear. Workforce transition planning and attention to local economic effects deserve consideration alongside efficiency metrics as these tools scale.

The Core Barrier: Implementation Capacity

The most significant barrier to realizing AI's potential in clinical care is not algorithm development but organizational capacity for implementation, governance, and monitoring. Systematic reviews of AI adoption barriers find that technical challenges, technological adoption capacity, and reliability and validation concerns collectively dominate, alongside workforce knowledge gaps and insufficient stakeholder trust.⁶⁸⁶⁹ Under-resourced HDOs face these challenges most acutely: they lack dedicated AI governance personnel, the data infrastructure for local validation, the connectivity and internal systems needed for basic AI readiness, and the organizational processes for lifecycle management.

HAIP's work directly addresses this implementation gap. The Practice Network is generating real-world implementation evidence from safety-net settings, including deployment of ambient AI scribes and operational AI tools alongside governance support. The HEAAL framework provides a standardized approach for fairness evaluation, addressing the reality that most hospitals lack any systematic method for bias assessment.⁷⁰ HAIP's 31 best practice guides provide the kind of practical implementation

⁶⁵Hassan M, et al. "Barriers to and facilitators of artificial intelligence adoption in health care: Scoping review." *JMIR Human Factors* (2024).

⁶⁶Hwang YM, et al. "The landscape of AI implementation in US hospitals." *Nature Health* (2026).

⁶⁷Gulamali FF, et al. "Eliminating the AI digital divide by building local capacity." *PLOS Digital Health* (2025).

⁶⁸Hassan M, et al. "Barriers to and facilitators of artificial intelligence adoption in health care: Scoping review." *JMIR Human Factors* (2024).

⁶⁹Katirai A. "A comprehensive overview of barriers and strategies for AI implementation in healthcare." *PLOS ONE* (2024).

⁷⁰Kim JY, et al. "Health Equity Across the AI Lifecycle (HEAAL)." *PLOS Digital Health* (2024).

frameworks that systematic reviews consistently identify as missing from the field.⁷¹ And the AI Vendor Disclosure Framework addresses information asymmetry in procurement, giving under-resourced organizations a structured way to evaluate vendor claims without building evaluation capacity from scratch.⁷²

Promising Tools and Federal Opportunities

The novel AI tools with the greatest potential to improve outcomes and reduce costs are not necessarily the most technically sophisticated. They are the ones that can be implemented effectively across diverse healthcare settings, with the organizational resources and infrastructure needed to adopt them responsibly.

AI tools that reduce administrative burden deserve continued federal attention. Ambient scribes have demonstrated the strongest evidence to date, but their long-term value will depend on whether business models evolve toward genuine efficiency rather than coding intensity. Operational AI for patient access, prior authorization, and revenue cycle management is delivering measurable gains, particularly for resource-constrained organizations. Independent evaluations that measure clinical outcomes and cost-effectiveness, not just vendor-reported metrics, would help organizations and payers distinguish effective products from those that primarily capture revenue.

Shared infrastructure for AI evaluation and monitoring would be transformative. No individual FQHC or community hospital can build the technical capacity to validate AI tools locally, yet local validation is essential for patient safety. Shared evaluation platforms, benchmarking tools, and monitoring systems would allow organizations of all sizes to assess AI on the merits, creating the conditions for a competitive market where better products win.

Capacity building for under-resourced settings addresses the most common reason AI tools fail to deliver value: organizations lack the people, processes, and infrastructure to implement them effectively. Technical assistance programs using hub-and-spoke approaches, where institutions with established expertise provide structured guidance to safety-net providers, can close this gap far more efficiently than expecting each organization to build capacity independently. These programs also need to address foundational AI readiness, including connectivity, internal systems, and workforce skill development, not just governance and workflow design. HAIP's Practice Network provides a demonstrated approach, and federal funding through grants and cooperative agreements could enable national scale.

Standardized vendor transparency would accelerate market efficiency. When organizations can compare AI products on a common basis through consistent disclosure of training data, performance metrics, fairness evaluations, and integration requirements, the market rewards quality and innovation rather than marketing spend. This particularly benefits smaller AI developers who build better products but lack the distribution advantages of dominant EHR vendors.

⁷¹Hasan A, et al. "Aligning AI principles and healthcare delivery organization best practices." *npj Digital Medicine* (2025).

⁷²Kpodzro S, et al. "A Collaborative Best Practice Guide for Promoting AI Vendor Transparency in Healthcare." *SSRN* (2025).

Finally, workforce planning deserves attention as AI adoption scales. The efficiency gains that make AI attractive, particularly in operational and administrative functions, can translate into job displacement if organizations optimize solely for cost reduction. Federal programs that support AI adoption could incorporate workforce transition considerations, particularly for rural and safety-net settings where the economic ripple effects of healthcare workforce changes are most significant.

HAIP is positioned to contribute to each of these areas: publishing Practice Network outcomes to add real-world evidence from the settings where it is most needed, serving as a multi-site research infrastructure for implementation studies, developing shared AI monitoring infrastructure, and scaling capacity building nationally.

Question 7: Decision Makers and Administrative Hurdles

Summary

- **No single role owns AI adoption decisions:** AI adoption requires coordination across executive leadership, including the CEO/COO, CMO/CMIO, CIO/CTO, CNO, and CFO, yet most organizations lack an established process for reconciling competing priorities into coherent decisions. Organizations with centralized decision processes and an executive sponsor tend to move faster and more consistently.
- **Frontline clinicians and their supervisors determine whether AI is actually used:** Leadership sets strategy, but adoption succeeds or fails at the point of care. Clinician buy-in depends on workflow integration, demonstrated patient benefit, and trust. Constructing the right incentive structures, from peer influence to performance recognition to awareness of patient impact, remains an underappreciated challenge.
- **In FQHCs, influence is distributed across multiple levels:** Health center leadership and boards, Health Center Controlled Networks (HCCNs), and external regulators and payers all shape AI adoption decisions, but most HCCNs currently lack the resources and governance capacity to lead AI adoption on behalf of their member centers.
- **Primary administrative hurdles are organizational, not technical:** Immature governance processes, lack of dedicated AI personnel, absence of standardized procurement criteria, resource constraints, and unresolved legal questions collectively slow adoption and disadvantage smaller organizations and innovative startups.

Who Influences AI Adoption

AI adoption in healthcare delivery organizations (HDOs) requires coordination across executive leadership, but no single role owns the decision, and that absence of clear authority is itself one of the primary administrative hurdles slowing private sector innovation. The CEO or COO sets strategic priorities and allocates resources. The CMO or CMIO evaluates clinical appropriateness and physician adoption. The CIO or CTO manages technical integration and infrastructure. The CNO determines nursing workflow impact. The CFO evaluates return on investment. Each brings legitimate concerns, yet in most organizations there is no established process for reconciling them into a coherent AI adoption decision.⁷³ HAIP's research, based on 89 interviews across its member organizations, found no clear ownership of AI lifecycle management activities, including decisions about whether to implement or decommission AI tools. Some organizations give that authority to budget holders; others favor centralized governance committees. Organizations with clear, centralized decision processes and an executive sponsor tend to move faster and more consistently, while those where authority is diffuse often stall or produce contradictory decisions across departments.

The emergence of dedicated AI leadership roles signals that some health systems recognize this gap. Cleveland Clinic appointed its first Chief AI Officer in 2024, and other organizations including HCA Healthcare and NewYork-Presbyterian have created dedicated AI leadership positions to consolidate strategy and governance. But these appointments remain the exception. Most organizations distribute AI responsibilities across existing roles, creating coordination overhead that slows adoption and increases the risk of inconsistent decisions.

However, executive leadership alone does not determine whether AI is adopted in practice. Frontline clinicians and their immediate supervisors are, in many respects, the most important influence on whether AI tools are actually used in clinical care. Leadership can set strategy, allocate resources, and build governance structures, but adoption succeeds or fails at the point of care. Clinician resistance driven by workflow disruption, alert fatigue, lack of trust, or insufficient training can render even well-governed AI tools ineffective.⁷⁴ Constructing incentive structures that drive frontline buy-in, whether through leadership engagement, peer influence, performance and career development, financial incentives, or awareness of patient impact, is an underappreciated challenge that organizational governance structures alone cannot solve. HAIP's Practice Network experience reinforces this: even with governance support and leadership commitment, the organizations seeing the strongest adoption are those that invest in clinician engagement and workflow co-design from the outset.

The picture is different, and more challenging, in the safety net. Within the FQHC system, influence is distributed across three levels: health center leadership and boards, who are responsible for clinical quality and fiduciary oversight but often lack the technical expertise to evaluate AI solutions; Health Center Controlled Networks (HCCNs), designated by HRSA to provide shared services and infrastructure, which function as de facto system-level decision makers for technology adoption; and

⁷³Kim JY, et al. "Organizational governance of emerging technologies: AI adoption in healthcare." *ACM FAccT* (2023).

⁷⁴Hassan M, et al. "Barriers to and facilitators of artificial intelligence adoption in health care: Scoping review." *JMIR Human Factors* (2024).

external regulators and payers, whose compliance and reimbursement requirements shape whether AI tools are viewed as feasible or risky to deploy. Despite their formal role in technology coordination, most HCCNs currently lack the dedicated resources, workforce capacity, and governance structures required to lead AI adoption responsibly on behalf of their member health centers, leaving individual centers to navigate these decisions largely on their own.

Primary Administrative Hurdles

The most pervasive administrative hurdle is that organizations are deploying AI while simultaneously trying to build governance structures for it. HAIP's research found that across multiple HDOs, clinical committees, IT departments, and compliance offices each held partial jurisdiction over AI, but none had complete oversight, and there was no consensus on how governance processes should work.⁷⁵ A systematic review of 35 AI governance frameworks confirmed that most existing frameworks were not designed for real-world application in complex healthcare environments.⁷⁶ The problem is compounded by the pace of change: AI capabilities are evolving so rapidly that static governance frameworks risk becoming outdated before they are fully implemented, requiring governance approaches that are adaptive by design rather than fixed.

The barriers fall into three reinforcing categories. First, process barriers: organizations lack standardized guidance on what to request from AI vendors, vendor responses are inconsistent, and there are no common procurement criteria. Without standardized procurement criteria, many HDOs default to the AI embedded in their existing EHR platform, not because it outperforms alternatives, but because the effort of evaluating independent products exceeds their available resources. This dynamic favors incumbent vendors over innovative startups developing better tools. Unclear regulation adds friction: liability allocation between vendors and HDOs remains unresolved, data use agreements are complex and variable, intellectual property disputes are common, and the absence of standard contract language for AI procurement means every deal is negotiated from scratch.

Second, people barriers: most HDOs lack dedicated AI governance personnel. IT and clinical staff are stretched thin across existing responsibilities, and the specialized expertise needed to evaluate AI tools, including algorithm performance, fairness assessment, and workflow integration, is in short supply. This expertise gap affects every stage of the AI lifecycle, from procurement through monitoring and decommissioning.

Third, resource barriers: budgets at most organizations cannot absorb the overhead of building governance infrastructure from scratch, particularly when the return on investment for governance itself is difficult to quantify. These costs are real but invisible: staff time diverted from clinical work, consultant fees for legal and technical review, and the opportunity cost of delayed adoption while governance structures mature.

⁷⁵Kim JY, et al. "Organizational governance of emerging technologies: AI adoption in healthcare." *ACM FAccT* (2023).

⁷⁶Hussein R, et al. "Advancing healthcare AI governance: A comprehensive maturity model based on systematic review." *npj Digital Medicine* (2026).

Each of these barriers represents duplicative effort across the healthcare system, with every HDO solving the same problems independently rather than building on shared solutions. For FQHCs, these barriers are structural: the absence of standardized governance frameworks leaves individual health centers to assess clinical risk, fairness, and accountability on their own; competitive rather than collaborative funding environments prevent system-level solutions; and misalignment between funding structures and shared services means the organizations serving the most complex patient populations are often the least equipped to govern AI responsibly.

Strategies That Overcome These Hurdles

Despite these challenges, a growing number of health systems have published their governance experiences, and they converge on a recognizable pattern. Organizations that successfully adopt AI, whether large academic medical centers like Memorial Sloan Kettering and Mass General Brigham, integrated systems like Kaiser Permanente, share several structural features.⁷⁷⁷⁸⁷⁹ They establish centralized, multidisciplinary governance committees that span clinical, technical, legal, and operational expertise rather than leaving AI decisions siloed in any one department. They create structured intake and assessment processes so that every new AI tool is evaluated against consistent criteria rather than ad hoc judgment. They take a lifecycle approach, governing AI from pre-deployment evaluation through ongoing monitoring and potential decommissioning. And they calibrate oversight to the level of risk, giving higher-risk clinical AI more intensive scrutiny while allowing lower-risk administrative tools to move through streamlined pathways.

The absence of these structures is precisely the administrative hurdle. Organizations without centralized committees see AI decisions stall in turf disputes between IT, clinical leadership, and compliance. Organizations without structured intake processes waste months in ad hoc evaluation cycles that exhaust staff time and produce inconsistent results. Organizations without lifecycle governance deploy tools and then discover performance problems only when harm has occurred. And organizations without tiered risk frameworks apply the same heavy process to low-risk scheduling tools and high-risk clinical decision support, creating unnecessary burden that slows adoption of both. Governance that is proportionate to organizational capacity and AI risk, with lighter-touch approaches for smaller or lower-risk settings, is more likely to be adopted and sustained.

HAIP has developed practical tools to make these governance structures accessible beyond the well-resourced institutions that pioneered them. The PPTO framework (People, Process, Technology, and Operations) defines the organizational roles, workflows, technology requirements, and operational procedures needed for AI lifecycle management. Originally developed through qualitative research across

⁷⁷Stetson PD, et al. "Responsible artificial intelligence governance in oncology." *npj Digital Medicine* (2025).

⁷⁸Tsoi AH, et al. "Establishing and implementing a responsible artificial intelligence framework: A 1-year review." *JAMIA* (2025).

⁷⁹Saenz AD, et al. "Establishing responsible use of AI guidelines: A comprehensive case study for healthcare institutions." *npj Digital Medicine* (2024).

HAIP's member organizations, the framework has been validated at a Canadian hospital system.^{80,81}

HAIP's 31 best practice guides, organized around 8 key decision points, translate governance principles into step-by-step procedures with identified stakeholders and data sources, covering both predictive AI (such as clinical risk scores) and generative AI (such as ambient documentation tools).⁸² HAIP is currently conducting a new wave of interviews with health system leaders across the U.S. to identify best practices for constructing governance structures, with the goal of making these practitioner-oriented insights available to the broader field and applying them in safety-net settings through the Practice Network.

Federal Opportunities

The most productive federal role is to disseminate and support the private sector frameworks that already work. Through ASTP/ONC and AHRQ, there is an opportunity to publish tiered implementation guidance and template documents, including intake surveys, assessment tools, and committee charters, so that each organization is not forced to reinvent governance from scratch. In the safety net, HRSA investment in governance capacity building at FQHCs and strengthening HCCNs as trusted intermediaries would reduce the duplicative burden that currently forces each health center to solve the same problems independently.

HAIP is positioned to support these efforts. The PPTO framework, best practice guides, and Practice Network collectively provide a ready-made foundation for governance capacity building across diverse settings. The consortium's ongoing research on governance structures at health systems nationwide is generating the kind of practical, practitioner-oriented insights that implementation guidance requires, and HAIP's Practice Network provides a demonstrated model for governance capacity building, with hub institutions providing structured guidance to safety-net organizations through a learning community.

⁸⁰Kim JY, et al. "Organizational governance of emerging technologies: AI adoption in healthcare." *ACM FAccT* (2023).

⁸¹Kim JY, et al. "People, process, technology, and operations framework for establishing AI governance in healthcare organizations." *npj Digital Medicine* (2026).

⁸²Hasan A, et al. "Aligning AI principles and healthcare delivery organization best practices." *npj Digital Medicine* (2025).

Question 8: Interoperability

Summary

- **Data standards for market competition:** Dominant EHR vendors control access to clinical data and workflow integration points, giving bundled AI tools a structural advantage regardless of quality. FHIR-based interoperability and continued USCDI expansion would break this bottleneck and let AI tools compete on performance. Broadening USCDI to include data elements critical for AI, such as the Medication Administration Record for inpatient models, and strengthening information blocking enforcement would open the market to independent developers, including startups building superior products.
- **Research data flows:** TEFCA does not currently include research as a permitted exchange purpose, forcing AI developers to negotiate bilateral data sharing agreements that effectively limit AI research to well-resourced institutions. Expanding TEFCA to include research would accelerate development, validation, and monitoring of AI across diverse populations.
- **Benchmarking that reflects real-world care:** Widely used benchmark datasets come disproportionately from academic medical centers and do not reflect the data quality, patient populations, or infrastructure of community hospitals, FQHCs, or rural facilities. Federated benchmarking approaches that include diverse care settings would generate evidence on how AI actually performs where it is needed most.

Data Standards That Enable Market Competition

The most impactful step toward widening market opportunities for clinical AI is to ensure that health data flows freely and in standardized formats across systems. Today, dominant EHR vendors control access to clinical data and workflow integration points, creating a structural advantage for AI tools bundled into their platforms, regardless of whether those tools are the best available. As discussed in our response to Question 2, enforcement of information blocking rules (45 CFR Part 171) has not kept pace with how this data control restricts competition. Independent AI developers, including startups building genuinely superior products, struggle to access the data they need to validate and deploy their tools in clinical settings. Enhanced interoperability would break this bottleneck, enabling a competitive market where AI tools win on performance rather than distribution.

FHIR (Fast Healthcare Interoperability Resources), HL7's modern API-based standard, provides the technical foundation for this vision and is the appropriate primary standard for AI-related data exchange. The United States Core Data for Interoperability (USCDI) establishes which data elements flow through FHIR-based APIs. ONC/ASTP has continued advancing USCDI adoption, and the HTI-5 Proposed Rule calls for USCDI version 3.1, which would expand the data classes and elements available through standardized APIs.⁸³ Future USCDI versions present an opportunity to prioritize the data elements most critical for AI development, including domains not yet well represented. The Medication Administration Record (MAR), for example, is particularly relevant for inpatient AI models but is not currently required as a standardized data element, limiting the ability of AI developers to build and validate tools across institutions.⁸⁴ Standardized data also enables what is currently difficult to achieve: fair comparative analysis of AI solutions across vendors, giving payers, providers, and patients the information they need to evaluate products on merit and driving quality-based competition across the market.

Beyond data access, there are contractual barriers that limit market competition for AI. Some EHR vendors impose requirements that third-party AI vendors seek approval before hiring employees who have had access to the EHR firm's proprietary information within the past 12 months. These restrictions limit independent AI developers' ability to recruit the technical expertise needed to build effective integrations, further entrenching the incumbents' advantage. Strengthening information blocking protections to address these contractual practices would help ensure that market competition reflects product quality rather than vendor lock-in.

Expanding Data Exchange to Fuel Research

The Trusted Exchange Framework and Common Agreement (TEFCA) establishes national infrastructure for health information exchange, but its current permitted purposes do not include research. Expanding TEFCA to include research as a permitted purpose for data exchange would significantly improve the data flows needed to develop, validate, and monitor clinical AI across diverse patient populations and care settings.⁸⁵ Without this expansion, researchers and AI developers must negotiate bilateral data

⁸³Health Data, Technology, and Interoperability: ASTP/ONC Deregulatory Actions to Unleash Prosperity (HTI-5 Proposed Rule), 90 Fed. Reg. 60970 (proposed Dec. 29, 2025).

⁸⁴ONC/ASTP. "United States Core Data for Interoperability (USCDI)." <https://isp.healthit.gov/united-states-core-data-interoperability-uscdi>

⁸⁵Sequoia Project. "TEFCA Exchange Purposes." <https://rce.sequoiaproject.org/exchange-purposes-explained/>

sharing agreements with individual health systems, a process so burdensome that it effectively limits AI research to well-resourced academic medical centers with existing data partnerships.

The data types where enhanced interoperability would most accelerate AI development span both structured and unstructured information. Medical imaging data, standardized through DICOM, already supports the largest category of FDA-cleared AI devices and represents a relative interoperability success story. Clinical notes, the unstructured text that captures the richness of clinical reasoning, are critical for the current generation of generative AI tools. Large language models can now process clinical text effectively without requiring extensive standardization beyond consistent use of English, a point that argues for prioritizing other data domains where standards are more urgently needed. Social determinants of health (SDOH) data remains poorly standardized and raises important questions about how such information is collected, shared, and used. As AI tools increasingly incorporate social context into clinical predictions, establishing minimum SDOH data standards through USCDI would improve both the fairness and accuracy of these tools, though careful attention to how this sensitive information is governed will be essential. Real-world outcomes data, including patient-reported outcomes and longitudinal follow-up information, is often missing from EHRs entirely, creating a fundamental gap in the evidence base for evaluating whether AI tools actually improve patient outcomes over time.

Benchmarking Tools That Reflect Real-World Care

Current benchmarking datasets for clinical AI skew heavily toward academic medical centers and do not reflect the diversity of care settings, patient populations, or data quality that AI tools encounter in real-world deployment. Widely used research datasets like MIMIC-IV (an EHR dataset covering emergency department and ICU admissions at Beth Israel Deaconess Medical Center) and the eICU Collaborative Research Database provide valuable resources for algorithm development, but their populations and data quality do not generalize to community hospitals, FQHCs, or rural facilities.⁸⁶ An AI tool validated against these benchmarks may perform well in a research paper and fail in a safety-net clinic.

There is an opportunity to address this gap through AHRQ and ONC/ASTP by supporting the development of benchmarking infrastructure that includes data from diverse care settings, particularly safety-net providers, rural facilities, and FQHCs whose patients are underrepresented in existing datasets. Rather than building massive centralized datasets, which raise privacy and governance concerns, federated benchmarking approaches that allow AI tools to be evaluated across sites without requiring data to leave local systems offer a more practical path forward. This would enable smaller healthcare organizations to participate in benchmarking without the infrastructure burden of large-scale data sharing, while generating evidence on how AI performs in the settings where it is needed most.

⁸⁶Johnson AEW, et al. "MIMIC-IV, a freely accessible electronic health record dataset." *Scientific Data* (2023). <https://doi.org/10.1038/s41597-022-01899-x>

Question 9: Patient and Caregiver Perspectives

Summary

- **Access and administrative burden:** Patients consistently identify access to care as the challenge they most want AI to address, with 30% of US adults citing improved access as AI's top expected benefit. Patients also feel the downstream effects of clinician burnout and administrative burden. In an AMA survey, 57% of physicians identified administrative burden automation as AI's biggest opportunity. AI tools that free clinician time for patient interaction, reduce wait times, and extend care to underserved areas align with patient priorities.
- **Trust and human oversight:** Patient acceptance depends on maintaining the human relationship. Only about 20% of US adults expect AI to improve their relationship with their doctor. Patients want transparency about how AI is used in their care, how their data is handled, and whether AI recommendations can be explained. The degree of transparency provided to patients can be proportionate to the clinical risk the AI tool presents.
- **Patients as partners:** Patient perspectives remain underrepresented in AI development and governance. Patient and provider priorities diverge: patients focus on barriers to accessing care, while providers focus on workflow efficiency. Meaningful patient engagement in problem identification, design, and monitoring would help ensure AI development serves the people it is meant to benefit.

What Patients Want AI to Fix

Patients and caregivers consistently identify access to care as the challenge they most want AI to address. Approximately 30% of US adults expect AI to improve their access to healthcare, the highest-rated expected benefit in recent surveys, with patients in underserved communities particularly prioritizing solutions that reduce wait times, enable remote monitoring, and extend specialty care to areas where it is currently unavailable.⁸⁷ AI also holds promise for better management of chronic disease through personalized health management systems. Approximately half of chronic disease patients view AI-enabled monitoring technologies as beneficial for their ongoing care.⁸⁸

Patients also feel the downstream effects of the administrative burden that consumes clinician time. In the AMA's 2024 physician survey, 57% of physicians identified administrative burden automation as AI's biggest opportunity, 75% said AI helps work efficiency, and 54% said it helps reduce burnout.⁸⁹ These are not just provider concerns: when physicians spend more time on documentation than on patient interaction, patients experience the consequences through shorter visits, delays in test results and follow-up communications, and fragmented coordination between providers. AI applications like ambient documentation tools directly target this problem by freeing clinician time for the patient relationship. There is also emerging potential for large language models to provide patients with basic information about test results directly, improving timely access to their own health information.

Beyond access and administrative burden, patients value AI's potential to improve diagnostic accuracy and timeliness, detecting conditions earlier through imaging analysis, reducing diagnostic errors, and providing second opinions on complex cases. At the same time, AI detection of subclinical disease that clinicians cannot yet correlate or mitigate raises questions about how to communicate such findings responsibly. Cost and affordability also matter: roughly 19% of US adults expect AI to make healthcare more affordable, though evidence on whether current AI tools actually reduce costs for patients remains limited.⁹⁰

Patient and Caregiver Concerns

Patient acceptance of AI is not unconditional, and trust is paramount. The most consistent concern across the research literature is that AI should assist rather than replace human clinicians.^{91,92} Only about 20% of US adults expect AI to improve their relationship with their doctor, and patients express strong preferences for human oversight in sensitive, complex, or long-term care situations.⁹³ Ensuring that

⁸⁷Nong P & Ji M. "Expectations of healthcare AI and the role of trust." *JAMIA* (2025).

⁸⁸Moy S, et al. "Patient perspectives on the use of AI in health care: A scoping review." *J Patient-Centered Research and Reviews* (2024).

⁸⁹American Medical Association. "Augmented Intelligence Research: Physician Sentiment Report." (February 2025). <https://www.ama-assn.org/system/files/physician-ai-sentiment-report.pdf>

⁹⁰Peterson Health Technology Institute. "Adoption of AI in healthcare delivery systems: Early applications and impacts." (2025). <https://phti.org/ai-adoption-early-applications-impacts/>

⁹¹Moy S, et al. "Patient perspectives on the use of AI in health care." *J Patient-Centered Research and Reviews* (2024).

⁹²Richardson JP, et al. "Patient apprehensions about the use of artificial intelligence in healthcare." *npj Digital Medicine* (2021).

⁹³Nong P & Ji M. "Expectations of healthcare AI and the role of trust." *JAMIA* (2025).

humans remain in the loop to verify that outcomes are accurate, explainable, and unbiased is central to maintaining the trust that effective care requires.

Data privacy ranks among the top concerns. Patients want to understand how their information is used in AI systems, whether data is appropriately protected, and who has access to AI-derived insights about their health. This concern takes on particular urgency as AI tools increasingly incorporate social determinants of health data, sensitive information about housing, food security, and economic circumstances that patients may share in one care setting without expecting it to flow across systems. As one of our community partners observed, this data is being collected repeatedly across different parts of the system, creating administrative burden and raising questions about whether the information is managed in the interest of the patient. Ensuring appropriate governance of this data, including updates to how protected health information requirements apply to AI use, is essential to maintaining trust.⁹⁴ Revising digital patient privacy rules has the potential to both strengthen patient protections and incentivize innovation through clearer pathways for data access.

Transparency is closely related. Patients are concerned about AI systems that cannot explain their recommendations, the "black box" problem, because lack of interpretability undermines informed decision-making and damages trust.⁹⁵ Patients also worry about AI perpetuating or worsening existing disparities: whether data-source bias affects recommendations, whether AI trained on limited populations will work for them, and whether AI will narrow or widen gaps in care quality. Patients in low-income communities and communities of color have a higher prevalence of chronic diseases and could benefit most from AI-assisted screening and diagnosis, but only if those tools are designed with appropriate fairness considerations.⁹⁶

Finally, patients want to know when AI is being used in their care. Rather than treating this as a binary question of informed consent, which refers to a specific legal process not feasible for every AI interaction, a more practical approach is progressive degrees of patient transparency proportionate to the model's clinical risk. Low-risk AI applications such as scheduling optimization may require only general disclosure, while AI that directly influences diagnostic or treatment decisions warrants more detailed communication about the tool's role relative to clinician judgment. The FDA has authorized over 1,250 AI-enabled medical devices as of mid-2025, yet a recent analysis found that many AI tools have entered clinical use without rigorous evaluation or meaningful public scrutiny, with critical details such as testing procedures, validation cohorts, and bias mitigation strategies often missing from regulatory documentation.^{97,98}

⁹⁴45 C.F.R. § 164.506(c)(5).

⁹⁵Fehr J, et al. "A trustworthy AI reality-check: The lack of transparency of AI products in healthcare." *Frontiers in Digital Health* (2024).

⁹⁶Dankwa-Mullan I. "Health Equity and Ethical Considerations in Using Artificial Intelligence in Public Health and Medicine." *Preventing Chronic Disease* (2024).

⁹⁷FDA. "Artificial Intelligence-Enabled Medical Devices." <https://www.fda.gov/medical-devices/software-medical-device-samd/artificial-intelligence-enabled-medical-devices>

⁹⁸Abulibdeh R, Celi LA, & Sejdic E. "The illusion of safety: A report to the FDA on AI healthcare product approvals." *PLOS Digital Health* (2025).

Patients as Partners, Not Just Recipients

A consistent finding across the research is that patient perspectives are underrepresented in AI development and governance.⁹⁹¹⁰⁰ Patients prioritize being engaged from the beginning, at problem identification and prioritization stages, not consulted after tools have already been built. This gap matters because patient and provider priorities differ: patients focus on barriers to accessing care, while providers tend to focus on efficiency and workflow. Research on what patients actually want from AI remains scarce despite rapid adoption, and without systematic patient input, the market for clinical AI risks optimizing for provider and vendor priorities rather than the patient outcomes that should ultimately drive value.

HAIP has built patient engagement into its governance structure from the outset. A community representative serves on HAIP's founding leadership team, bringing connections to grassroots coalitions including the Comunidad de Durham network that engage communities on AI and health advocacy. The HEAAL framework explicitly identifies patient and community stakeholders for each of its 37 assessment procedures, making patient involvement a structural feature of AI evaluation rather than an afterthought.¹⁰¹ The Practice Network extends this commitment by working with FQHCs and community hospitals that serve populations whose voices are least represented in AI development.

There are several opportunities for HHS to strengthen patient engagement in AI through existing authorities. AHRQ could fund research on patient priorities for AI across diverse communities, building the evidence base that the field currently lacks. CMS could incorporate patient experience measures into AI-related quality programs, ensuring that patient outcomes, not just technical performance, drive adoption incentives. And ONC/ASTP could include patient-facing transparency requirements in future rulemaking, giving patients the information they need to make informed decisions about AI in their care.

⁹⁹Moy S, et al. "Patient perspectives on the use of AI in health care." *J Patient-Centered Research and Reviews* (2024).

¹⁰⁰Adus S, et al. "Exploring patient perspectives on how they can and should be engaged in AI development." *BMC Health Services Research* (2023).

¹⁰¹Kim JY, et al. "Development and preliminary testing of Health Equity Across the AI Lifecycle (HEAAL): A framework for healthcare delivery organizations to mitigate the risk of AI solutions worsening health inequities." *PLOS Digital Health* (2024).

Question 10: Research Priorities

Summary

- **Implementation science, not algorithm development:** Despite thousands of AI models validated in research settings, translation into clinical practice has been remarkably slow. The evidence base is strongest for diagnostic accuracy and workflow efficiency, but evidence on whether AI actually improves patient health outcomes, reduces costs, or narrows disparities remains sparse. Federal research funding continues to favor algorithm development over the implementation science that would close this gap.
- **Costs, benefits, and value:** Even for ambient AI, the most mature generative AI application, the Peterson Health Technology Institute found that evidence on financial performance is uncertain. Implementation costs are consistently understudied. Who captures the value from AI efficiency gains is poorly understood, and the return on investment extends beyond financial metrics to include clinician well-being and workforce retention.
- **Priority areas:** HHS is uniquely positioned to fund implementation science research, close the capacity gap between well-resourced and safety-net providers, support AI tools that expand the healthcare workforce, invest in shared evidence infrastructure, and develop research methods for emerging applications like agentic AI systems.

Published Evidence: What Exists and What Is Missing

Published evidence on AI's clinical impact has grown substantially, though critical gaps remain. The strongest evidence exists for diagnostic accuracy, where controlled studies show AI often rivals clinicians in tasks like medical image interpretation, and for workflow efficiency, where ambient AI documentation tools have demonstrated clinician time savings.¹⁰² The Peterson Health Technology Institute's evaluation of major health systems found that ambient scribes likely reduce clinician burnout, though financial impact remains unclear.¹⁰³ Approximately half of chronic disease patients view AI-enabled monitoring technologies as beneficial for their ongoing care, and there is growing evidence that AI can improve chronic disease management through personalized health management systems.¹⁰⁴

However, this evidence comes disproportionately from academic medical centers and does not generalize well to safety-net settings where data quality, patient populations, and technical infrastructure differ significantly. Evidence on clinical outcomes, whether AI actually improves patient health, remains sparse, with few randomized controlled trials measuring patient outcomes rather than algorithm performance metrics. A 2024 study of AI randomized controlled trials found that only about 15% explicitly cited the CONSORT-AI reporting guidelines, and much of the available data has been acquired retrospectively, making it difficult to draw causal conclusions about AI's impact on care delivery.¹⁰⁵¹⁰⁶ Evidence on downstream effects of workflow tools, whether time saved translates into more patient interaction, reduced burnout, or improved throughput, also remains limited. Most published research evaluates algorithm accuracy metrics rather than the outcomes that matter to patients, providers, and the healthcare system: improved health, reduced cost, better access, and enhanced clinician well-being.

Costs, Benefits, and Who Captures Value

The literature on AI costs and benefits remains immature. Even for ambient AI, the PHTI evaluation found that evidence on productivity and financial performance is uncertain, with health system leaders reporting gaps in evidence regarding impact on the number of patients providers can see or the accuracy of billing codes.¹⁰⁷ Implementation costs are consistently understudied: the substantial resources required for local validation, governance, workforce training, and ongoing monitoring are rarely quantified, yet these costs often determine whether an AI tool is feasible for a given organization. Initial implementation costs remain particularly prohibitive for safety-net settings, where the return on investment must be demonstrated before budgets can absorb the upfront burden.¹⁰⁸

¹⁰²Rajpurkar P, et al. "AI in health and medicine." *Nature Medicine* 28(1): 31-38 (2022).

¹⁰³Peterson Health Technology Institute. "Adoption of AI in healthcare delivery systems." (2025). <https://phti.org/ai-adoption-early-applications-impacts/>

¹⁰⁴Moy S, et al. "Patient perspectives on the use of AI in health care: A scoping review." *J Patient-Centered Research and Reviews* (2024).

¹⁰⁵CDA-AMC (Canada's Drug Agency). "2025 Watch List: Artificial Intelligence in Health Care." (2025). <https://www.cda-amc.ca/2025-watch-list-artificial-intelligence>

¹⁰⁶Plana D, et al. "Concordance of randomised controlled trials for artificial intelligence interventions with the CONSORT-AI reporting guidelines." *Nature Communications* (2024).

¹⁰⁷Peterson Health Technology Institute. "Adoption of AI in healthcare delivery systems." (2025). <https://phti.org/ai-adoption-early-applications-impacts/>

¹⁰⁸Yousefi F, et al. "Opportunities, challenges, and requirements for AI in primary health care." *BMC Primary Care* (2025).

A key question the literature has not adequately addressed is how the value from AI efficiency gains is distributed across stakeholders, including patients, clinicians, health systems, and payers. There is substantial anecdotal evidence about the potential for AI to create value across the care delivery system, but very little rigorous evidence on how that value is captured by different stakeholders. One area where the dynamics are becoming clearer is ambient AI scribes, where a recent policy brief documented how vendor marketing is shifting from burnout reduction to revenue-cycle performance, with business models increasingly focused on more intensive coding rather than cost reduction for patients.¹⁰⁹ This raises questions about whether the current market is directing AI innovation toward genuine value creation.

The return on investment for AI also extends beyond financial metrics. Decreased clinician burnout, improved staff retention, and enhanced workforce satisfaction are real benefits that take time to materialize and are rarely measured in current cost-benefit analyses, yet they may ultimately prove more important to sustainable AI adoption than short-term productivity gains. At the same time, cost reduction through AI-driven automation of functions like call centers and revenue cycle management raises legitimate concerns about workforce displacement, particularly in rural communities where healthcare organizations are often a primary employer.

Priority Research Areas

The most critical gap in the AI evidence base is not algorithm development but implementation science. The field has shifted from internal validation to external validation and deployment, yet federal research funding continues to favor algorithm development.¹¹⁰ HHS is uniquely positioned to correct this imbalance, and doing so would accelerate the private sector's ability to bring effective AI tools to patients.

First, implementation science research through AHRQ, NIH, and CMS would address the most urgent gap, studying how to integrate AI into clinical workflows, manage organizational change, and sustain adoption across diverse care settings. Pragmatic multicenter trials that evaluate AI in real-world conditions, rather than controlled research environments, would generate the evidence that healthcare delivery organizations need to make informed adoption decisions. This includes research on effective AI governance structures for organizations of different sizes and resources, ongoing monitoring approaches for detecting performance drift and emerging risks, decommissioning criteria for when and how underperforming AI tools should be retired, and methods for assessing the behavioral changes AI introduces in clinical decision-making.

Second, research on closing the capacity gap between well-resourced health systems and safety-net providers would help ensure that AI benefits reach all patients. Studies on effective technical assistance models, including hub-and-spoke approaches where resource-rich institutions support community providers, would generate evidence on how to scale AI adoption beyond the academic medical centers that currently dominate the field. Research on AI readiness, particularly connectivity and infrastructure

¹⁰⁹Dai T, Kvedar JC, & Polsky D. "Policy brief: Ambient AI scribes and the coding arms race." *npj Digital Medicine* (2025).

¹¹⁰CDA-AMC (Canada's Drug Agency). "2025 Watch List: Artificial Intelligence in Health Care." (2025). <https://www.cda-amc.ca/2025-watch-list-artificial-intelligence>

requirements in rural settings, is essential to ensuring that federal investments in AI reach the communities that stand to benefit most. FDA and ARPA-H also have roles to play in supporting translational research that bridges the gap between validated algorithms and deployable clinical tools.

Third, research on AI tools that expand the reach of the healthcare workforce deserves priority, particularly tools supporting community health workers and home-based caregivers. International evidence demonstrates that community health workers can cost-effectively manage chronic disease, maternal health, and preventive care, yet in the US their impact is often constrained by administrative burden and limited clinical scope. Research on AI tools that reduce documentation burden for frontline workers, enable safe use of clinical decision support in community settings, and develop appropriate training and governance models for non-physician AI use could significantly extend care capacity without requiring new workforce pipelines.

Fourth, the field needs shared infrastructure for generating and synthesizing real-world evidence. Standardized reporting frameworks building on CONSORT-AI and SPIRIT-AI, shared benchmarking tools that enable cross-site performance comparison, and federated data approaches that support evidence generation without requiring centralized data repositories would all reduce the burden on individual organizations while producing evidence that reflects the diversity of care delivery in the United States.

Fifth, emerging AI capabilities require new research methods. Agentic AI systems that can take autonomous actions in clinical workflows, rather than simply providing recommendations, present novel questions about impact assessment, downstream propagation of errors, and appropriate governance. Research on systematic methods for evaluating these more autonomous AI applications, before they reach widespread clinical deployment, would help the field stay ahead of the technology curve rather than reacting after problems emerge.